

Robust visual tracking with extreme point graph-guided annotation: Approach and experiment

Peng Gao^{a,g,*}, Xin-Yue Zhang^a, Xiao-Li Yang^a, Feng Gao^b, Hamido Fujita^{c,d,e}, Fei Wang^f

^a School of Cyber Science and Engineering, Qufu Normal University, Qufu, Shandong 273165, China

^b School of Computer Science and Technology, East China Normal University, Shanghai 200062, China

^c Malaysia–Japan International Institute of Technology (MJIT), Universiti Teknologi Malaysia, Kuala Lumpur 54100, Malaysia

^d Andalusian Research Institute in Data Science and Computational Intelligence (DaSCI), University of Granada, Granada 18011, Spain

^e Regional Research Center, Iwate Prefectural University, Takizawa, Iwate 020-0693, Japan

^f School of Electronics and Information Engineering, Harbin Institute of Technology, Shenzhen, Guangdong 518055, China

^g Yuntian Educational Group, Qingyun, Shandong 253700, China

ARTICLE INFO

Keywords:

Visual tracking

Siamese networks

Representation learning

Keypoint detection

ABSTRACT

Recent advancements in the field of visual tracking have been propelled by the amalgamation of Siamese networks and region proposal networks, which have demonstrated excellent competitive accuracy while remaining computationally efficient. However, these approaches often suffer from excessive parametric redundancy and additional computational costs owing to the use of anchor boxes or multiscale pyramids, leaving room for improvement in their performance. In this study, a novel extreme point graph-guided tracking approach called SiamEXTR is presented, which tracks generic target objects by detecting five keypoints, including four target-specific extreme points (i.e., the topmost, bottommost, leftmost, and rightmost points) and a center point, without requiring region classification or bounding box regression. To enhance the robustness of the extreme point detection, a new oriented pooling strategy called extreme-pooling is proposed, which captures more recognizable discriminative global and local information to help a pixel predict its category. In addition, a U-shaped backbone network is designed to preserve fine-grained visual details and stronger semantic information at high-resolution, ensuring that the detection granularity of the extreme point graph is as close to the subpixel-level as possible. Based on the detected extreme point graph, the proposed approach not only generates axis-aligned bounding boxes for object annotation, but also provides more accurate octagonal object segmentation masks through a simple approximation strategy. Without bells and whistles, extensive experiments and comparisons on several authoritative large-scale benchmark datasets demonstrated that the SiamEXTR tracker consistently achieved competitive performance, with running speeds significantly exceeding 140 frames per second. The authors hope that the concept behind this approach will serve as a new baseline and promote further development of the visual tracking community.

1. Introduction

Visual tracking is a fundamental research topic in the field of computer vision (Li, Wang, Wang and Lu, 2018; Smeulders et al., 2014; Thangavel, Kokul, Ramanan, & Fernando, 2023; Yang, Shao, Zheng, Wang, & Song, 2011). It plays a vital role in a variety of real-world applications, such as video surveillance, intelligent vehicles, human–computer interaction, and robotics, and has recently attracted tremendous attention from various communities (Javed et al., 2022; Marvasti-Zadeh, Cheng, Ghanei-Yakhdan, & Kasaei, 2022; Ondrašovič & Tarábek, 2021). Visual tracking approaches attempt to estimate the

location and size of an arbitrary target object throughout a video sequence, which is specified by a bounding box only in the first frame.

Despite making massive progress in visual tracking over the past decade (Bertinetto et al., 2016; Chen, Peng, Wang, Lu, & Hu, 2023; Cui et al., 2022; Cui, Jiang, Wang and Wu, 2022; Gao, Ma, Song et al., 2018; Li, Yan, Wu, Zhu and Hu, 2018; Wu et al., 2023; Xing, Evangeliou, Tsoukalas, & Tzes, 2022; Ye, Chang, Ma, Shan, & Chen, 2022; Yu, Xiong, Huang, & Scott, 2020), developing effective and efficient tracking approaches remain a challenge because of numerous interfering factors, including illumination variations, fast motions, scale

* Corresponding author at: School of Cyber Science and Engineering, Qufu Normal University, Qufu, Shandong 273165, China.

E-mail addresses: pgao@qfnu.edu.cn (P. Gao), leo_zxy0727@foxmail.com (X.-Y. Zhang), sdylx12547@163.com (X.-L. Yang), fgao@stu.ecnu.edu.cn (F. Gao), hufujita-799@acm.org (H. Fujita), wangfeiz@hit.edu.cn (F. Wang).

<https://doi.org/10.1016/j.eswa.2023.122013>

Received 7 February 2023; Received in revised form 24 September 2023; Accepted 2 October 2023

Available online 6 October 2023

0957-4174/© 2023 Elsevier Ltd. All rights reserved.

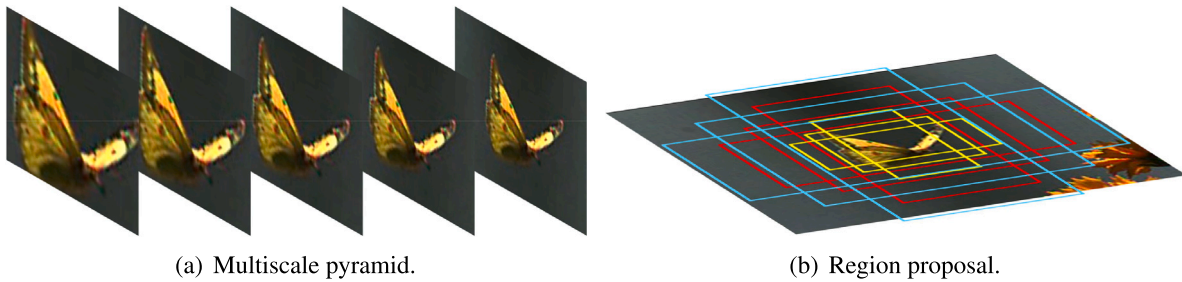


Fig. 1. Visualization of two dominating scale estimation paradigms. For the sake of clarity, we construct a five-scale pyramid in (a), and nine anchor boxes around the target object location using three scales and three aspect-ratios in (b). The example frame is shown from *butterfly* sequence (Kristan et al., 2018).

variations, occlusions, and deformations (Fan et al., 2021; Huang, Zhao, & Huang, 2019; Kristan et al., 2018; Mueller, Smith, & Ghanem, 2016; Wu, Lim, & Yang, 2015).

With the rapid renaissance of deep learning techniques (LeCun, Bengio, & Hinton, 2015; Schmidhuber, 2015), convolutional neural networks (CNNs) have revolutionized several essential research areas in computer vision, leading to extraordinary performance improvements in image recognition (Krizhevsky, Sutskever, & Hinton, 2012; Simonyan & Zisserman, 2015), object detection (Lin et al., 2017; Ren, He, Girshick, & Sun, 2017), human pose estimation (Fieraru, Khoreva, Pishchulin, & Schiele, 2018; Newell, Yang, & Deng, 2016), and semantic segmentation (Badrinarayanan, Kendall, & Cipolla, 2017; Li, Xiong, An and Wang, 2018). The past few years have witnessed significant advances in generic visual tracking, partly driven by the introduction of deep learning, enabling remarkable progression over time (Marvasti-Zadeh et al., 2022). Several pioneering approaches (Danelljan, Häger, Khan, & Felsberg, 2015; Danelljan, Robinson, Khan, & Felsberg, 2016; Ma, Huang, Yang, & Yang, 2019; Qi et al., 2019) in the visual tracking community have attempted to integrate CNN features instead of conventional handcrafted features (Felzenszwalb, Girshick, McAllester, & Ramanan, 2010; Van De Weijer, Schmid, Verbeek, & Larlus, 2009) with a successful discriminative correlation filter (DCF) framework (Javed et al., 2022) to obtain record-breaking performance. Recently, Siamese trackers have drawn considerable attention from the visual tracking community owing to their high efficiency and effectiveness (Bertinetto et al., 2016; Cui, Guo et al., 2022; Gao, Yuan et al., 2020; Li, Yan et al., 2018; Zhang, Peng, Fu, Li, & Hu, 2020). These trackers formulate visual tracking as a matching problem and train a Siamese network (SIAMNET) to predict a similarity score map between the CNN features of the initial object exemplar and the search candidate in the current frame. The target object is then tracked to the location where the highest similarity score is obtained. Often, SIAMNET is trained offline on large-scale video object detection datasets, and the network hyperparameters are fixed during online tracking. With powerful CNN features and an efficient SIAMNET framework, this type of tracker has achieved favorable results in recent challenging visual tracking competitions (Kristan et al., 2020, 2019). However, an intractable obstacle in the aforementioned approaches lies in handling variations in the scale of the target object; thus, much work is still required to enhance the accuracy and reliability of visual tracking.

Currently, two dominant strategies are adopted in most visual tracking approaches to mitigate the impact of scale variations. The first strategy involves detecting the target object location, typically the geometric center, based on DCF or SIAMNET, followed by leveraging an exhaustive multiscale image or feature pyramid (Danelljan, Häger, Khan and Felsberg, 2017; Li, Chen, Wang and Zhang, 2019) centered around the detected location, as shown in Fig. 1(a). The changes in target object sizes can then be estimated by searching for the heatmap with the highest similarity score across all scales (Bertinetto et al., 2016; Gao, Ma, Song et al., 2018; Gao, Ma, Yuan, Xiao, & Wang, 2019). The second strategy involves generating anchor boxes (Girshick, 2015; Ren et al., 2017) densely over the input image based on preset

hyperparameters, including the number, scales, and aspect-ratios, and then performing joint classification and regression using region proposal networks (RPN) to determine the optimal target object location and size simultaneously (Fan & Ling, 2019; Li, Yan et al., 2018), as shown in Fig. 1(b). The former strategy is much simpler but severely harmful to the tracking speed (Danelljan, Bhat, Khan and Felsberg, 2017; Danelljan, Bhat, Khan, & Felsberg, 2019), whereas the latter has a desirable inference speed, but involves expensive training and requires a large number of complex manually designed parameters; for example, the RPN module in SiamRPN (Li, Yan et al., 2018) contains more than 20 million hyperparameters. In addition, these strategies have limited adaptability to scale, resulting in difficulties in accurately estimating the scale. Additionally, both strategies use axis-aligned bounding boxes as annotations that tightly encompasses the target object. Nevertheless, the shapes of most target objects are not rectangular (generally, most target objects are non-rigid bodies), and the axis-aligned bounding boxes are only approximations, making it inevitably includes several background distractors to cover the entire target object. Therefore, the tracking performance cannot be guaranteed when background features are enclosed. In such cases, these strategies are not feasible for practical applications.

To address the aforementioned issues, SiamEXTR, a novel fully-convolutional approach that technically formulates generic visual tracking as an extreme point graph detection problem to enhance tracking accuracy and robustness, is proposed in this study. SiamEXTR follows the SIAMNET framework (Bertinetto et al., 2016; Gao, Zhang et al., 2020; Li, Yan et al., 2018) and predicts a set of detection heatmaps by comparing the CNN features of a given object exemplar to those of a much larger search candidate in a sliding-window fashion. The location with the highest score on each heatmap corresponds to an extreme point. In our opinion, a satisfactory tracking approach is required to localize the target objects accurately and estimate their sizes robustly. In the extreme point graphs, four target-specific extreme points are used to annotate the target objects, as shown in Fig. 8. The center point serves to locate the target object and enforces an extra geometric constraint between the opposite pairs of extreme points during training. In other words, a pair of diagonal points (Gao, Yuan et al., 2020; Yang et al., 2021), including a top-right point and bottom-left point, can be inferred based on four detected extreme points. Further, these two points should be symmetric around the center point.

However, the use of vanilla CNN features alone may not be sufficient for extreme point graph detection because these features often lack the necessary spatial visual evidence of the target object at sharp boundaries (Fan & Ling, 2019; Li, Wu et al., 2019). To address this issue, a novel strategy called extreme-pooling is introduced, which employs a series of oriented max-pooling operations to extract global and local target-specific information at potential extreme point locations. Moreover, the detection of extreme point graphs for visual tracking presents a significant challenge owing to the requirement for both high-resolution and fine-grained contextual information. Several pre-trained CNN models for image classification and detection exhibit decreasing spatial resolution as the architecture becomes deeper, which

is not conducive to subpixel-level detection. Considering this limitation, a U-shaped backbone network is proposed that captures both fine-grained geometric details and stronger semantic information at multiple resolutions and propagates them to a higher-resolution in a recursive manner, similar to U-Net (Ronneberger, Fischer, & Brox, 2015) architectures. This design enables accurate detection of extreme point graphs, particularly for small target objects.

SiamEXTR is a fully-convolutional anchor-free Siamese tracking approach that utilizes a subpixel-level extreme point graph detection scheme. This approach involves training the network offline and subsequently using it for online tracking. Without bells and whistles, empirical experiments demonstrated that SiamEXTR achieves superior tracking performance in terms of both robustness and speed, making it a promising approach for real-time tracking applications. The key contributions of this study are as follows:

- A fully-convolutional anchor-free Siamese tracker for the generic visual tracking task, which is formulated as an extreme point graph detection problem.
- A novel feature pooling strategy enriches the representation of extreme points by leveraging both global and local target-specific information.
- A U-shaped backbone network enables the feature maps to be yielded at high-resolution and maintains the most robust informative representations.

The remainder of this paper is organized as follows. Section 2 provides a brief review of the recent advances related to the study. Section 3 elaborates the methodology used in the proposed approach. Section 4 details the experiments and discusses the results. Finally, Section 5 concludes the study.

2. Related works

In this section, the representative approaches for visual tracking that are most closely related to the current study are briefly reviewed. For more detailed descriptions and comparisons of state-of-the-art visual tracking methods, please refer to the most recent comprehensive reviews (Javed et al., 2022; Marvasti-Zadeh et al., 2022; Ondrašovič & Tarábek, 2021; Thangavel et al., 2023).

As an emerging force in the revolutionary trend of deep learning, Siamese trackers exhibit a promising balance between accuracy, robustness, and speed. SIAMNET is a fully-convolutional network architecture trained to compare two input image patches and determine their similarity (Bromley, Guyon, LeCun, Säckinger, & Shah, 1993). In the realm of visual tracking, SIAMNET is trained to compare the object exemplar in the initial frame of a video sequence with the search candidates in subsequent frames in a dense sliding-window manner to estimate the location and size of the given target object. Two pioneering Siamese trackers, SINT (Tao, Gavves, & Smeulders, 2016) and SiamFC (Bertinetto et al., 2016), leveraged the SIAMNET architecture for visual tracking leaving an indelible mark on the field and shaping subsequent developments. One of the paramount challenges facing Siamese trackers is managing scale variations. The size of the target object may fluctuate considerably, posing difficulties for trackers, particularly for those that are diminutive or massive.

An extensively adopted strategy for combating the effects of scale variation in visual tracking is the use of a multiscale pyramid search approach. This strategy has been employed in both handcrafted-feature-based (Danelljan, Häger et al., 2017; Gao, Ma, Li et al., 2018; Li & Zhu, 2014) and deep-feature-based (Gao, Ma, Song et al., 2018; Gao, Zhang et al., 2020; Tao et al., 2016) tracking approaches. For instance, SiamFC (Bertinetto et al., 2016) incorporates a search at five different scales, i.e., $1.035^{[-2, -1.0, 1.2]}$, and linearly interpolates the size of the target object by a factor of 0.35 to achieve a smooth transition. However, RAR (Gao, Zhang et al., 2020) only utilizes three scales, that

is $1.02^{[-1.0, 1]}$, which strikes an acceptable balance between the tracking performance and speed. Various SIAMNET-based tracking approaches such as RASNet (Wang et al., 2018), StructSiam (Zhang et al., 2018), FlowTrack (Zhu, Wu, Zou, and Yan, 2018), TADT (Li, Ma, Wu, He and Yang, 2019), SA-Siam (He, Luo, Tian, & Zeng, 2018), GradNet (Li, Chen, Ouyang et al., 2019), and RTINET (Yao, Wu, Zhang, Shan, & Zuo, 2018) also benefit from this scale estimation strategy. However, there is no consensus on the optimal number of scale candidates to search for before the returns diminish. Regardless, the multiscale pyramid strategy incurs additional computational costs, irrespective of speed, and cannot be trained end-to-end. Consequently, the current trend diverges away from this scale estimation strategy.

Another line of efforts to deal with scale variations is the anchor-based strategy, which endeavors to deploy an RPN (Ren et al., 2017) integrated with SIAMNET (Bertinetto et al., 2016; Tao et al., 2016) to yield accurate visual tracking results (Fan & Ling, 2019; Li, Yan et al., 2018; Yu et al., 2020; Zhang & Peng, 2019; Zhu, Wang et al., 2018). The anchor-based strategy was first introduced in the visual tracking community by SiamRPN (Li, Yan et al., 2018), which consists of two branches for separately estimating the location and size of the target object. Similar to object detection (Girshick, 2015; Girshick, Donahue, Darrell, & Malik, 2014; Ren et al., 2017), SiamRPN utilizes preset anchor boxes with a wide variety of scales and aspect-ratios at each pixel-level location. The classification branch estimates the foreground and background of each anchor box, and the regression branch is employed simultaneously to deduce the adjustment hyperparameters. Subsequently, several extensions have been proposed to significantly enhance tracking performance through the implementation of a multi-stage cascaded RPN in C-RPN (Fan & Ling, 2019), class-agnostic binary segmentation in SiamMask (Wang, Zhang, Bertinetto, Hu and Torr, 2019), multilevel RPN in SiamRPN++ (Li, Wu et al., 2019), and region refinement module in SiamAttn (Yu et al., 2020). Several recent state-of-the-art tracking approaches such as DaSiamRPN (Zhu, Wang et al., 2018), SiamDW (Zhang & Peng, 2019), SPLT (Yan, Zhao, Wang, Lu, & Yang, 2019), CSA (Yan, Wang, Lu, & Yang, 2020), SPM (Wang, Luo, Xiong and Zeng, 2019), and SiamCAN (Zhou et al., 2021) have also adopted this strategy. Although, the anchor-based strategy avoids multiscale pyramid search using region proposals and advances the SIAMNET-based tracking framework, it still requires the design and implementation of preset anchor boxes as well as the tuning of a multitude of additional hyperparameters to achieve optimal tracking performance, which inevitably augments the computational complexity.

In the last two years, there has been a growing interest in using the anchor-free strategies to address issues with scale variations. This strategy is more efficient and flexible than anchor-based strategies because it can estimate the location and size of the target object without relying on preset anchor boxes. Several studies demonstrated the effectiveness of this strategy. For instance, SATIN (Gao, Yuan et al., 2020) and SiamCorners (Yang et al., 2021) utilize a keypoint-aware anchor-free strategy (Law & Deng, 2018) that first identifies a set of preset keypoints and then performs bounding box regression based on these keypoints. SiamCAR (Cui, Guo et al., 2022) and SiamGAT (Guo et al., 2021) employ a center-ness anchor-free strategy (Zhu, He, & Savvides, 2019) to classify the foreground and background of each pixel-level location and regress the distance from the search candidates to the ground-truth target object center point. Recently, certain Siamese trackers, such as SiamFC++ (Xu, Wang, Li, Yuan, & Yu, 2020), Ocean (Zhang et al., 2020), SiamBAN (Chen, Zhong, Li, Zhang, & Ji, 2020), and SiamRCR (Peng et al., 2021), have also exploited this strategy for scale estimation and achieved a tracking performance comparable to that of anchor-based Siamese trackers. The authors believe that the anchor-free strategy has the potential to improve the accuracy and robustness of visual tracking and will continue to be an active of research area.

In this study, a novel anchor-free Siamese tracker inspired by successful object detection methods such as ExtremeNet (Zhou, Zhuo, & Krahenbuhl, 2019) and CenterNet (Duan et al., 2019) is introduced.

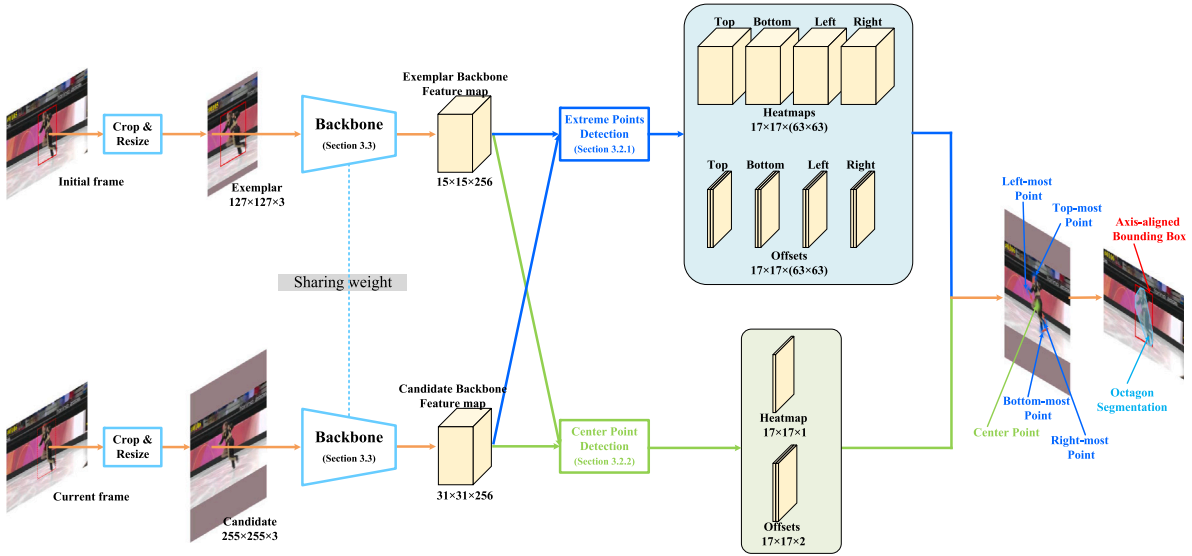


Fig. 2. Overview of the proposed extreme point graph-guided tracking approach.

Unlike previous scale estimation strategies that rely on anchor boxes, the use of extreme point graph detection is proposed to eliminate the need for anchor box-related hyperparameters. The proposed tracker utilizes a fully-convolutional SIAMNET framework and can be trained end-to-end. This study aims to achieve exceptional tracking performance while maintaining a stable real-time capability.

3. Proposed approach: SiamEXTR

This section details how the proposed SiamEXTR utilizes the extreme point graph of generic target objects to guide the visual tracking process. Fig. 2 shows the overall pipeline of SiamEXTR. This approach begins by extracting backbone feature maps from an object exemplar and a search candidate, which are cropped from the initial and current frames of the video sequence, respectively. Subsequently, an extreme point graph is detected on the search backbone feature map. Based on the probability score of each pixel being an extreme point, the optimal locations of the four extreme points are selected and refined using their corresponding offsets. Finally, an axis-aligned bounding box or octagon segmentation mask is generated based on the extreme point graph to annotate the location and size of the target object. For clarity, first, the classical SIAMNET-based tracking framework is revisited in Section 3.1, followed by a description of how to formulate the visual tracking problem as an extreme point graph detection task in Section 3.2. The architecture of the proposed U-shaped backbone network is detailed in Section 3.3. Finally, the multi-task tracking loss and tracking annotation strategies are presented in Sections 3.4 and 3.5, respectively.

3.1. SiamNet framework revisited

SIAMNET-based tracking approaches utilize object exemplar z and search candidate x to estimate the location and size of a target object over time. The object exemplar is obtained according to the ground-truth bounding box in the initial frame, whereas the search candidate is cropped from subsequent frames based on the location and size of the target object estimated in the last frame. Next, a heatmap g is obtained by a cross-correlation operation (Bertinetto et al., 2016) in a sliding-window manner as follows:

$$g(z, x) = F(z) \star F(x) \quad (1)$$

where $\star$ stands for the cross-correlation operation, and F indicates the backbone feature maps. The location of the target object in the

current frame is determined by the highest score on g . Subsequently, the multiscale pyramid search, anchor-based, or anchor-free strategies mentioned in Section 2 are employed to estimate the size of the target object. Finally, the target object is annotated using axis-aligned bounding boxes in the current frame.

3.2. Extreme point graph detection

The proposed SiamEXTR utilizes extreme point graphs to provide robust and accurate annotations of the target object, which can be accomplished by detecting the target-specific topmost, bottommost, leftmost, rightmost, and center points.

3.2.1. Extreme points detection

The topmost point is considered as an example of formulating a target object as an extreme point. In general, the target-specific topmost point refers to the highest vertical point on the target object. In the formulation process, the backbone feature maps are integrated with global and local visual evidence at the extreme point locations using an extreme-pooling strategy, which comprises a series of oriented max-pooling operations. For convenience, let $F \in \mathbb{R}^{W \times H \times C}$ denote the backbone feature map, here W , H , and C indicate the width, height, and dimension (a.k.a., the number of channels) of F , respectively. The coordinates of the top-left and bottom-right pixel points of F along the spatial axis are denoted by $(1, 1)$ and (W, H) , respectively. For a pixel point $\{(m, n, d) \mid m \in \{1, \dots, W\}, n \in \{1, \dots, H\}, d \in \{1, \dots, C\}\}$ on F , a spatial max-pooling operation is conducted from the bottom-most side towards the top-most side of the backbone feature map (a.k.a., the “Top Pooling” in Fig. 3) as follows:

$$F_p^t(m, n, d) = \max(F(m, n, d), F(m, n+1, d), \dots, F(m, H-1, d), F(m, H, d)) \quad (2)$$

where F_p^t is the output of F after the vertical upward max-pooling operation. Subsequently, the max-pooling operations are respectively performed from the rightmost side and the leftmost sides along the horizontal direction to the leftmost side and the rightmost side as follows:

$$F_p^{t,l}(m, n, d) = \max(F_p^t(m, n, d), F_p^t(m+1, n, d), \dots, F_p^t(W-1, n, d), F_p^t(W, n, d)) \quad (3)$$

$$F_p^{t,r}(m, n, d) = \max(F_p^t(1, n, d), F_p^t(2, n, d), \dots, F_p^t(m-1, n, d), F_p^t(m, n, d)) \quad (4)$$

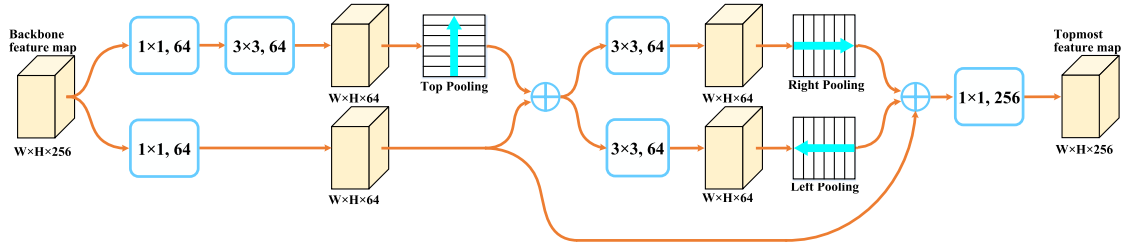
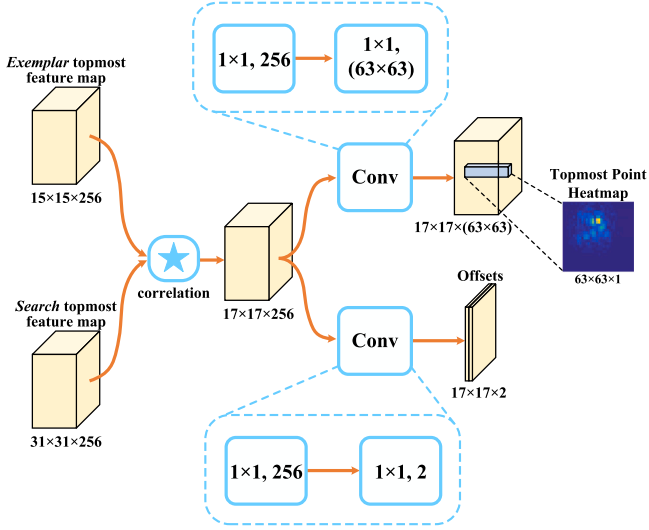


Fig. 3. Illustrate of the topmost extreme-pooling process.

Fig. 4. The pipeline for the topmost point detection. The symbol $\star$ denotes cross-correlation.

where $\mathbf{F}_p^{t,l}$ and $\mathbf{F}_p^{t,r}$ are the output of $\mathbf{F}_p^t$ after the horizontal leftward and rightward max-pooling operations (a.k.a., the “Left Pooling” and “Right Pooling” in Fig. 3), respectively. The topmost feature map $\mathbf{F}_t$ of the target object can be obtained by directly adding $\mathbf{F}_p^{t,l}$ and $\mathbf{F}_p^{t,r}$. Similarly, the target-specific bottommost, leftmost, and rightmost feature maps, which are based on specific extreme-pooling strategies, are denoted as $\mathbf{F}_b$, $\mathbf{F}_l$ and $\mathbf{F}_r$, respectively.

It is worth noting that during the extreme-pooling operation, the dimensions of the backbone feature map are firstly compressed by 1/4 using a 1×1 convolutional layer to reduce the computation overhead. At the end of the process, the extreme point feature map is raised to the same dimensions as the backbone feature map using another 1×1 convolutional layer, as shown in Fig. 3.

Without loss of generality, let $\mathbf{F}_t(\mathbf{z}) \in \mathbb{R}^{W_z \times H_z \times C}$ and $\mathbf{F}_t(\mathbf{x}) \in \mathbb{R}^{W_x \times H_x \times C}$ denote the topmost feature maps of the object exemplar and the search candidate, respectively. Then, a heatmap can be generated by cross-correlating $\mathbf{F}_t(\mathbf{z})$ and $\mathbf{F}_t(\mathbf{x})$ as follows:

$$\mathbf{g}_t(\mathbf{z}, \mathbf{x}) = \mathbf{F}_t(\mathbf{z}) \star \mathbf{F}_t(\mathbf{x}) \quad (5)$$

where $\mathbf{g}_t(\mathbf{z}, \mathbf{x}) \in \mathbb{R}^{W_t \times H_t \times C}$, $W_t = W_x - W_z + 1$ and $H_t = H_x - H_z + 1$.

Subsequently, a detection heatmap $\mathbf{S}_t$ and an offset map $\mathbf{O}_t$ can be obtained based on $\mathbf{g}_t$ using two different branches, including an extreme point detection branch $\varphi_{s,t}$ and an offset estimation branch $\varphi_{o,t}$, respectively, as shown in Fig. 4. The extreme point detection branch consists of a 1×1 convolutional layer with output dimension C and a 1×1 convolutional layer with output dimension $W_t \times H_t$, and the offset estimation branch comprises a 1×1 convolutional layer with output dimension C and a convolutional layer with output dimension 2.

$$\begin{aligned} \mathbf{S}_t &= \varphi_{s,t}(\mathbf{g}_t(\mathbf{z}, \mathbf{x})) \\ \mathbf{O}_t &= \varphi_{o,t}(\mathbf{g}_t(\mathbf{z}, \mathbf{x})) \end{aligned} \quad (6)$$

The vector $\mathbf{S}_t^\lambda \in \mathbb{R}^{1 \times 1 \times (W_t \times H_t)}$ in $\mathbf{S}_t$ represents the probability of each pixel being the topmost extreme point, which lies in the λ -th sliding window with the same size as $\mathbf{F}_t(\mathbf{z})$ on $\mathbf{F}_t(\mathbf{x})$, that means $\mathbf{S}_t^\lambda = \varphi_{s,t}(\mathbf{g}_t(\mathbf{z}, \mathbf{x}))$. $\mathbf{O}_t$ is a two-channel offset map used to adjust the estimated extreme point locations, where one channel adjusts the x-axis coordinates, and the other adjusts the y-axis coordinates. All vectors $\{\mathbf{S}_t^\lambda \mid \lambda \in \{1, 2, \dots, W_t \times H_t\}\}$ of the detection heatmap share the same $\mathbf{O}_t$. The location of the target-specific topmost point in the current frame can be determined by the maximum score on the detection heatmap $\mathbf{S}_t$, and is finally adjusted according to $\mathbf{O}_t$ (Gao, Yuan et al., 2020; Law & Deng, 2018).

3.2.2. Center point detection

The formulation and detection processes of the target center point are only employed in the offline training phase. In contrast, center point detection can guide the backbone network to learn more efficient visual representations of the target object based on the detected center point location. However, it can also enforce two geometric constraints between the topmost and bottommost points, as well as the leftmost and rightmost points.

For center point detection, formulating the target object as a center point is proposed using a series of oriented max-pooling operations, as shown in Fig. 5. The backbone network feature $\mathbf{F} \in \mathbb{R}^{W \times H \times C}$ is fed into four max-pooling branches, respectively. Specifically, these branches perform upward, downward, leftward and rightward max-pooling operations on $\mathbf{F}$, respectively (a.k.a., the “Top Pooling”, “Bottom Pooling”, “Left Pooling” and “Right Pooling” in Fig. 5), and the output features of the four oriented max-pooling operations are summed to obtain an informative representation of the center point. This is because the center points are often located inside the target object and are rich in semantic information for classification but lack sufficient geometric visual details. Using oriented max-pooling operations in each direction, the geometric visual details of the target object can be effectively fused with the semantic information inside it.

Let $\mathbf{F}_c(\mathbf{z}) \in \mathbb{R}^{W_z \times H_z \times C}$ and $\mathbf{F}_c(\mathbf{x}) \in \mathbb{R}^{W_x \times H_x \times C}$ denote the center feature maps of the object exemplar $\mathbf{z}$ and the searching candidate $\mathbf{x}$, respectively, and the center point probability map can be obtained according to the cross-correlation operation of Eq. (1),

$$\mathbf{g}_c(\mathbf{z}, \mathbf{x}) = \mathbf{F}_c(\mathbf{z}) \star \mathbf{F}_c(\mathbf{x}) \quad (7)$$

where $\mathbf{g}_c(\mathbf{z}, \mathbf{x}) \in \mathbb{R}^{W_c \times H_c \times C}$, $W_c = W_x - W_z + 1$ and $H_c = H_x - H_z + 1$.

Next, detection heatmap $\mathbf{S}_c$ and offset map $\mathbf{O}_c$ are produced using the location detection branch $\varphi_{s,c}$ and an offset estimation branch $\varphi_{o,c}$, respectively.

$$\begin{aligned} \mathbf{S}_c &= \varphi_{s,c}(\mathbf{g}_c(\mathbf{z}, \mathbf{x})) \\ \mathbf{O}_c &= \varphi_{o,c}(\mathbf{g}_c(\mathbf{z}, \mathbf{x})) \end{aligned} \quad (8)$$

where $\varphi_{s,c}$ consists of a 1×1 convolutional layer with output channel C and a 1×1 convolutional layer with output channel 1. Yet, the pipeline of $\varphi_{o,c}$ here is the same as that of the extreme point offset branch, as depicted in Fig. 6. This is because the center point is located inside the target object, and its detection process searches for the sliding window region that is most similar to $\mathbf{x}$ on $\mathbf{z}$. The center location of the target object in the current frame is determined by the maximum score on $\mathbf{S}^c$ and adjusted according to $\mathbf{O}^c$, too.

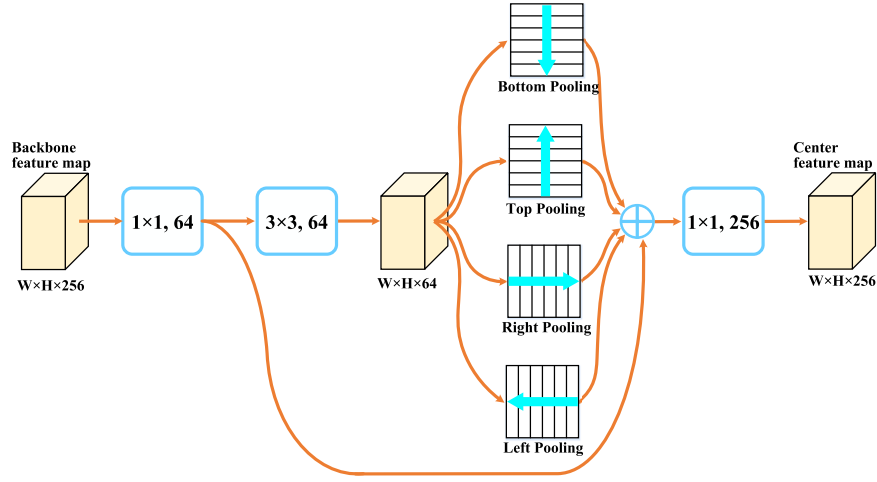


Fig. 5. Illustrate of the center point pooling process.

Table 1

Illustration of the U-shaped backbone network. More details of each layers in each residual convolutional block are shown in square parenthesis. The output sizes with and without round parenthesis indicate that of the searching candidate feature map and the object exemplar feature map, respectively.

Stages	Primary network			Refinement network		
	Output size	Residual conv blocks	Stride	Output size	Residual conv blocks	Stride
Input	255 × 255 (127 × 127)	–	–	–	–	–
1	127 × 127 (63 × 63)	[7 × 7, 64]	2	–	–	–
2	63 × 63 (31 × 31)	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$	4	31 × 31 (15 × 15)	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 3$	8
3	31 × 31 (15 × 15)	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 4$	8	15 × 15 (7 × 7)	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 4$	16
4	15 × 15 (7 × 7)	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 6$	16	7 × 7 (3 × 3)	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 6$	32
5	7 × 7 (3 × 3)	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$	32	3 × 3 (1 × 1)	$\begin{bmatrix} 1 \times 1, 1024 \\ 1 \times 1, 4096 \end{bmatrix} \times 3$	64
6	15 × 15 (7 × 7)	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 2$	16	7 × 7 (3 × 3)	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 2$	32
7	31 × 31 (15 × 15)	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 2$	8	15 × 15 (7 × 7)	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 1$	16
Output	31 × 31 (15 × 15)	[1 × 1, 256]	8	–	–	–
FLOPs (G)				9.25		
# parameters (M)				8.85		

3.3. U-shaped backbone network

Extreme point detection is a spatially sensitive subpixel-level detection problem. Most existing CNN models pre-trained for image classification learn feature maps that decrease in resolution as they become deeper in the network. Consequently, these CNNs are not well suited for subpixel-level detection that require fine-grained informative representations at high-resolution. Although, upsampling low-resolution feature maps may partially compensate for the loss of visual details, they are generally insufficient to achieve a robust performance in extreme point detection.

Therefore, SiamEXTR employs U-Net (Ronneberger et al., 2015) as its primary backbone architecture. U-Net is a well-known and simple neural network model that can effectively incorporate both low- and high-resolution feature maps by using shortcuts that connect the

encoding and decoding layers. These shortcut connections allow the network to make full use of both global semantic information from low-resolution feature maps and local visual details from high-resolution feature maps, which is important for efficiently presenting the target objects.

The backbone network in the proposed model inherits the U-shaped architecture of the U-Net, which has two parts: an encoder subnetwork and a decoder subnetwork. The encoder subnetwork is designed based on the fully convolutional part of ResNet-50 and is divided into five stages (a.k.a., “stage1” to “stage5” in Fig. 7) according to the resolutions of the feature maps. The resolution of the input image was reduced by half to at least 1/32 of the original-resolution when passing from one stage to the next. In contrast, the decoder subnetwork is made up of several upsampling layers and residual convolution blocks. It has two stages (a.k.a., “stage6” to “stage7” in Fig. 7), which gradually

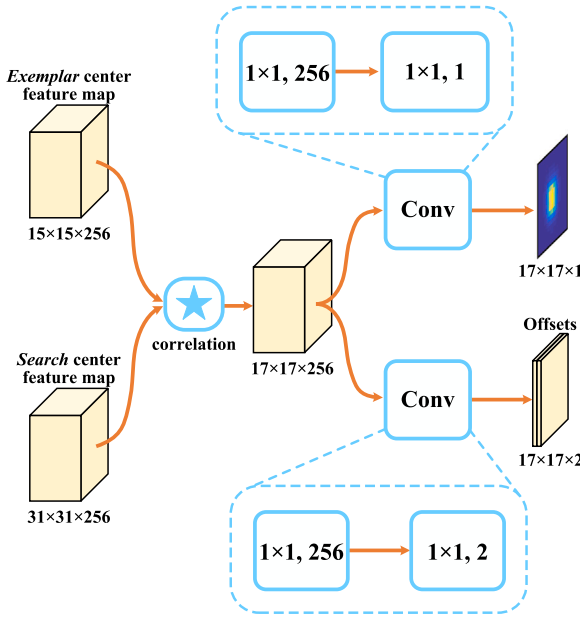


Fig. 6. The pipeline for the center point detection. The symbol $\star$ denotes cross-correlation.

doubles the resolution of the feature maps yielded from the encoder subnetwork when passing from one stage to the next, back up to 1/8 of the original-resolution. Similar to U-Net, the backbone network also includes two shortcuts that connect stages 3 and 7, as well as stages 4 and 6, allowing for the merging of lower-resolution semantic information with higher-resolution visual details and taking the lost fine-grained image context back to the backbone feature maps, which enables the gradient to propagate directly to the previous stages.

However, it can be argued that the aforementioned network based on ResNet-50, which will be further denoted as the “primary network”, is undesirable because of insufficient semantic capability. Therefore, an auxiliary feature refinement network denoted as the “refinement network” is incorporated into the primary network at stages 2 through 7 (six stages in total) enhance the overall semantic capability of high-resolution output feature maps. Each stage comprises several residual convolutional blocks.

It is noteworthy that the resolution of each stage in the refinement network is half that of the resolution of the corresponding stage of the primary network; however, the feature dimension is doubled. During the forward pass, the output feature maps from these blocks are fused with their corresponding feature maps from the primary network, as shown in Fig. 7. The goal of this fusion strategy is to use the semantic information from low-resolution feature maps to refine the target-specific visual details from high-resolution feature maps. Specifically, when the fused feature maps are entered in to the next residual convolutional block in the refinement network, the resolution of the primary output feature maps is first reduced using a 3×3 convolutional layer with a stride of 2, resulting in downsampled primary feature maps. Subsequently, the dimensions of the downsampled primary feature maps and refinement feature maps are adjusted using two 1×1 convolutional layers to 1/2 and 1/4, respectively. The resulting feature maps are then concatenated along the channel dimension and passed through a 1×1 convolutional layer to obtain the input feature maps for the next residual convolution block in the refinement network (a.k.a., the “Feature Fusion (Downsampling)” process in Fig. 7). When the fused feature maps are entered in to the next residual convolution block in the primary network, the nearest neighbor interpolation method is utilized to upsample the resolution of the refinement feature maps, following the same steps as before (a.k.a., the “Feature Fusion (Upsampling)” process in Fig. 7).

The proposed U-shaped backbone network can aggregate rich contextual representations in parallel between low- and high-resolution feature maps, which avoids the loss of fine-grained visual details and enables the exchange of the strongest semantic information across different resolutions. Moreover, this enables SiamEXTR to accurately detect extreme point maps of larger and smaller target objects, particularly for subpixel-level detection problems. Further details of the U-shaped backbone network are provided in Table 1.

3.4. Multi-task tracking loss

During the offline training process, if the positive and negative samples are labeled with the binary label $y \in \{0, 1\}$, then, only the ground-truth extreme points or center points are positive. Simultaneously, all other pixel points are negative. To increase the number of positive samples and solve the sample imbalance problem, the extreme point graph is labeled using soft labels with decay radius r as introduced in SATIN (Gao, Yuan et al., 2020). Let $(\hat{m}_t, \hat{n}_t)$ denote the ground-truth location of the topmost point in search candidate $\mathbf{x}$. For each pixel point (m, n) on $\mathbf{x}$, a soft label is defined as follows:

$$y_{(m,n),t} = \exp\left(-\frac{(m - \hat{m}_t)^2 + (n - \hat{n}_t)^2}{2\xi^2}\right) \quad (9)$$

where $\xi = r/3$ denotes the decay coefficient. The smaller ξ is, the faster $y_{(m,n),t}$ decay. The soft labels y_b , y_l , y_r and y_c correspond to the bottommost point $(\hat{m}_b, \hat{n}_b)$, leftmost point $(\hat{m}_l, \hat{n}_l)$, rightmost point $(\hat{m}_r, \hat{n}_r)$, and center point $(\hat{m}_c, \hat{n}_c)$ of the target object, respectively, and are generated in the same manner as in Eq. (9).

As described in Section 3.2, the extreme points are determined in each sliding window region of the same size as the object exemplar $\mathbf{z}$ on search candidate $\mathbf{x}$. Based on the dimensions of the exemplar feature maps defined in Eq. (5), $\Lambda = W_t \times H_t$ sliding window regions of size $W_t \times H_t$ can be generated on $\mathbf{x}$. Similarly, a sliding window label $y_{\lambda,t}$ corresponding to each of the Λ sliding window regions can be produced on the soft label y_t . Then, for the λ -th sliding window region, the location loss of the four extreme points can be expressed as follows:

$$\begin{aligned} \mathcal{L}_{ep,\lambda} = & - \sum_{i \in \{t,b,l,r\}} \frac{1}{W_t H_t} \sum_{m=1}^{W_t} \sum_{n=1}^{H_t} y_{\lambda,i}^{(m,n)} (1 - s_{\lambda,i}^{(m,n)}) \log(s_{\lambda,i}^{(m,n)}) \\ & + s_{\lambda,i}^{(m,n)} (1 - y_{\lambda,i}^{(m,n)}) \log(1 - s_{\lambda,i}^{(m,n)}) \end{aligned} \quad (10)$$

where $y_{\lambda,i}^{(m,n)}$ and $s_{\lambda,i}^{(m,n)}$ are the detection scores of the pixel point (m, n) on $y_{\lambda,i}$ and $S_{\lambda,i}$, respectively.

For all Λ sliding window regions on $\mathbf{x}$, the extreme points location loss can be formulated as follows:

$$\mathcal{L}_{ep} = \frac{1}{\Lambda} \sum_{\lambda} \mathcal{L}_{ep,\lambda} \quad (11)$$

The center point location loss can be expressed as follows:

$$\begin{aligned} \mathcal{L}_{cp} = & - \frac{1}{W_c H_c} \sum_{m=1}^{W_c} \sum_{n=1}^{H_c} y_c^{(m,n)} (1 - s_c^{(m,n)}) \log(s_c^{(m,n)}) \\ & + s_c^{(m,n)} (1 - y_c^{(m,n)}) \log(1 - s_c^{(m,n)}) \end{aligned} \quad (12)$$

where $y_c^{(m,n)}$ and $s_c^{(m,n)}$ are the detection scores of the pixel point (m, n) on y_c and S_c , respectively.

It is noteworthy that as the backbone network becomes deeper, the size of $\mathbf{F}(\mathbf{x})$ is usually smaller than that of $\mathbf{x}$, a discretization error occurs when the location of the ground-truth extreme point $(\hat{m}_t, \hat{n}_t)$ on $\mathbf{x}$ is mapped to the location $(\lfloor \hat{m}_t/\alpha \rfloor, \lfloor \hat{n}_t/\alpha \rfloor)$ in S_c , where α is the downsampling factor of the backbone network, and $\lfloor \cdot \rfloor$ denotes the round down operation. Thus, the extreme point graph remapping precision losses are defined as follows:

$$\hat{\mathbf{O}}_i = \left(\frac{\hat{m}_i}{\alpha} - \left\lfloor \frac{\hat{m}_i}{\alpha} \right\rfloor, \frac{\hat{n}_i}{\alpha} - \left\lfloor \frac{\hat{n}_i}{\alpha} \right\rfloor \right), \quad i \in \{t, b, l, r, c\} \quad (13)$$

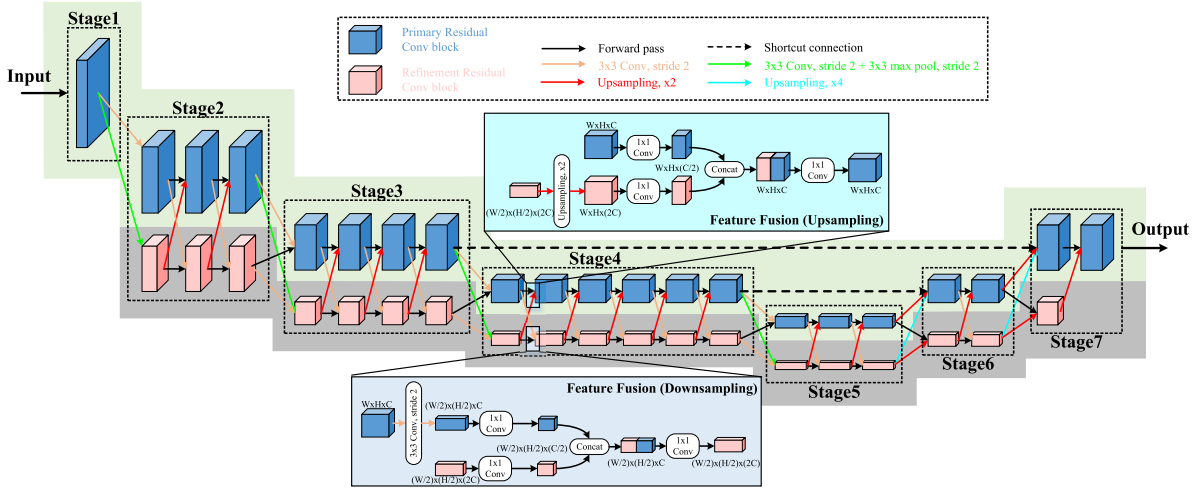


Fig. 7. Architecture of the proposed U-shaped backbone network.

Thereby, the offset loss can be formulated by employing the smooth $\mathcal{L}_1$ Loss (Girshick, 2015) as follows:

$$\mathcal{L}_{os} = \sum_{i \in \{t, b, l, r, c\}} \sum_{k=1}^2 \text{smooth}_{\mathcal{L}_1}(\mathbf{o}_i^k - \hat{\mathbf{o}}_i^k) \quad (14)$$

Additionally, an auxiliary center point $(\frac{m_l+m_r}{2}, \frac{n_l+n_b}{2})$ can be derived from the four detected extreme points: (m_l, n_l) , (m_b, n_b) , (m_r, n_r) and (m_r, n_r) . It is noteworthy that this auxiliary center point is not the detected center point (m_c, n_c) or the ground-truth center point $(\hat{m}_c, \hat{n}_c)$. The symmetric loss of the four detected extreme points is defined as follows:

$$\mathcal{L}_{sym} = \left(\frac{m_l + m_r}{2} - \hat{m}_c \right)^2 + \left(\frac{n_l + n_b}{2} - \hat{n}_c \right)^2 \quad (15)$$

Here, $\mathcal{L}_{sym}$ will be larger if $(\frac{m_l+m_r}{2}, \frac{n_l+n_b}{2})$ deviates far from the ground-truth center point $(\hat{m}_c, \hat{n}_c)$. The vertical distances from the detected topmost and bottommost points to the ground-truth center point are constrained to be equal by $\mathcal{L}_{sym}$, and the horizontal distances from the detected leftmost and rightmost points to the ground-truth center point should also be equal to achieve a more robust tracking performance. It is worth noting that the same selection strategy proposed by SiamRPN (Li, Yan et al., 2018) is utilized to suppress severe variations in the size and ratio of the detected result; more details are provided in the original paper of SiamRPN.

SiamEXTR is trained offline by minimizing a multi-task loss $\mathcal{L}$, including the extreme points location loss $\mathcal{L}_{ep}$, the center point location loss $\mathcal{L}_{cp}$, the offset loss $\mathcal{L}_{os}$, and the symmetric loss $\mathcal{L}_{sym}$, expressed as follows:

$$\mathcal{L} = \mathcal{L}_{ep} + \lambda_1 \mathcal{L}_{cp} + \lambda_2 \mathcal{L}_{os} + \lambda_3 \mathcal{L}_{sym} \quad (16)$$

where λ_1 , λ_2 and λ_3 are hyperparameters to control the balance of each loss terms.

3.5. Tracking annotation strategy

During the online tracking phase, the center point detection pipeline is discarded and only use the detected extreme point graph is used to form the axis-aligned bounding boxes that can more accurately annotate the target objects. Compared with the aforementioned multi-scale pyramid search, anchor-based, and existing anchor-free strategies, the proposed approach is carefully designed to mimic the recognition mechanisms of the human visual system, as it can accommodate the scale variations of target objects well.

In addition, the object segmentation strategy proposed in DEXTR (Maninis, Caelles, Pont-Tuset, & Van Gool, 2018) can be combined

with SiamEXTR to convert the detected extreme point graphs into more robust octagonal masks, as shown in Fig. 8.

4. Experiments

In this section, the experimental results of the proposed SiamEXTR tracker on six popular benchmark datasets are presented and compared with state-of-the-art trackers. Ablative analyses are conducted to investigate the efficiency and effectiveness of the components of the proposed tracking approach.

4.1. Experimental methodology

4.1.1. Implementation details

The proposed SiamEXTR is implemented in Python with PyTorch on an instance with 2 Intel® Xeon® E5-2698 v4 CPU @ 2.2 GHz CPU with 240 GB RAM, and 4 NVIDIA® Tesla® V100 GPU with 128 GB VRAM. Similar to SiamFC (Bertinetto et al., 2016) and SiamRPN (Li, Yan et al., 2018), the input sizes of the object exemplar and search candidate were set to 127×127 and 255×255 pixels, respectively.

During the offline training phase, the batch size was set to 128, and a total of 25 epochs were performed using stochastic gradient descent (SGD) with a first warm-up phase, in which the learning rate increased linearly from 10^{-3} to 5×10^{-3} for the first five epochs and then decreased logarithmically until 5×10^{-4} for 20 more epochs. SiamEXTR was trained using data from PASCAL (Everingham et al., 2015), COCO (Lin et al., 2014), DAVIS-2016 (Perazzi et al., 2016), DAVIS-2017 (Pont-Tuset et al., 2017), and YouTube-VOS (Xu et al., 2018). There are no direct extreme point graph-guided annotations in the aforementioned training datasets; however, these datasets are complete annotations of video object segmentation masks. Thus, the ground-truth can be obtained by determining the extreme points in the polygonal mask annotations (Maninis et al., 2018; Zhou et al., 2019). The U-shaped backbone network was pre-trained on the ImageNet-DET dataset (Russakovsky et al., 2015). Further, data augmentation for the object exemplar and search candidate was considered, which consists of random translation (up to 10 pixels), random rotation (up to 30°), random resizing (between $\times 0.8$ and $\times 1.2$), and random flipping.

All parameters were fixed across the experiments and datasets. SiamEXTR runs at around 142 frames per second (fps) on a single NVIDIA® Tesla® V100 GPU with 32 GB VRAM, which exceeds the real-time bound (Yang et al., 2011) by a huge margin.

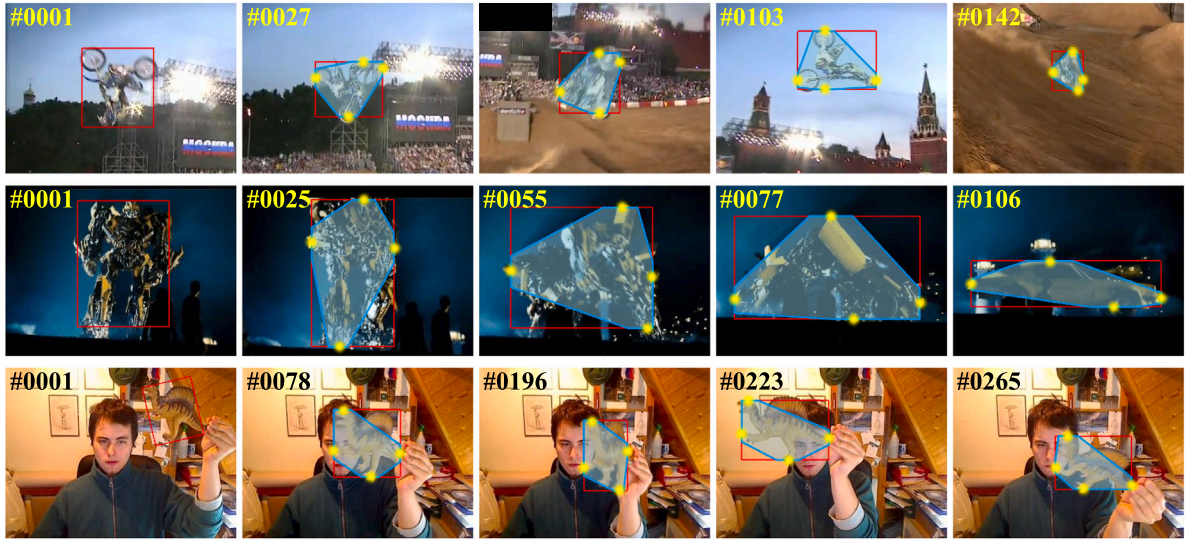


Fig. 8. Illustration of results of our proposed extreme point graph-guided tracking approach. It aims to trace target objects by detecting their extreme point graphs (in yellow), which not only form axis-aligned bounding boxes (in red) but also produce octagonal segmentation masks (in cyan) to annotate target objects more accurately. Example frames are shown from *MotorRolling* (top row), *Trans* (middle row) and *dinosaur* (bottom row) sequences.

Table 2

Details of different visual tracking benchmark datasets used in our experimental comparison.

Benchmark dataset	OTB100	VOT2016	VOT2018	UAV123	LaSOT	GOT-10k
Video sequences	100	60	60	123	1.4k	10k
Total frames	58.61k	21.455k	21.356k	113.476k	3.52M	1.5M
Minimum frames	71	24	24	109	1000	51
Maximum frames	3872	1507	1500	3085	11397	920
Average resolution	356 × 530	757 × 480	758 × 465	1231 × 699	632 × 1089	929 × 1638
Frame rate	30 fps	30 fps	30 fps	30 fps	30 fps	10 fps

4.1.2. Benchmark datasets

Six benchmark datasets, OTB100 (Wu et al., 2015), VOT2016 (Kristan, Leonardis et al., 2016), VOT2018 (Kristan et al., 2018), UAV123 (Mueller et al., 2016), LaSOT (Fan et al., 2021), and GOT-10k (Huang et al., 2019) were used to assess the performance of SiamEXTR. The details of these datasets are summarized in Table 2.

4.1.3. Evaluation metrics

To compare the performance of the trackers, benchmark datasets have utilized different evaluation metrics to evaluate the efficiency and effectiveness in terms of robustness, accuracy, and speed.

Distance precision (DP) measures the Euclidean distance between the estimated bounding box location (B_x, B_y) and the ground-truth annotation location (G_x, G_y) as follows:

$$DP = \|(B_x, B_y) - (G_x, G_y)\|_2 \quad (17)$$

The trackers are ranked using this metric with a threshold of 20 pixels (Mueller et al., 2016; Wu et al., 2015).

Normalized distance precision (DP_{Norm}) normalizes DP over the size of the ground-truth annotation G as follows:

$$DP_{Norm} = \|W((B_x, B_y) - (G_x, G_y))\|_2 \quad (18)$$

where W denotes the size of the ground-truth annotation (Fan et al., 2021).

Overlap success rate (OSR) measures the overlap ratio between the estimated bounding box B and ground-truth annotation G , which is defined as follows:

$$\text{Overlap ratio} = \frac{|B \cap G|}{|B \cup G|} \quad (19)$$

The trackers were evaluated using the area under the curve (AUC) of the OSR within an overlap ratio threshold range of 0 to 1 (Mueller et al., 2016; Wu et al., 2015).

Accuracy (A) and **average overlap (AO)** estimate the average overlap between the estimated bounding boxes B and ground-truth annotations G (Čehovin, Leonardis, & Kristan, 2016; Huang et al., 2019).

Robustness (R) measures the average failure times during tracking (Čehovin et al., 2016).

Expected average overlap (EAO) reflects both the A and R. For further definitions of this metric, please refer to Čehovin et al. (2016) and Kristan, Matas et al. (2016).

$SR_{0.50}$ and $SR_{0.75}$ denote the percentage of tracked frames where the overlap ratio exceeds a threshold of 0.50 and 0.75, respectively (Huang et al., 2019).

4.2. Results on OTB100

Experiments were conducted using the OTB100 (Wu et al., 2015) benchmark dataset. Following the standard protocol for the OTB benchmark datasets, the DP and AUC of the OSR were adopted to rank different trackers. A comparison of 15 state-of-the-art trackers including SiamCAR (Cui, Guo et al., 2022), SiamTPN (Xing et al., 2022), HiFT (Cao, Fu, Ye, Li, & Li, 2021), SiamCorners (Yang et al., 2021), Ocean (Zhang et al., 2020), RAR (Gao, Zhang et al., 2020), SATIN (Gao, Yuan et al., 2020), SiamAttn (Yu et al., 2020), SiamBAN (Chen et al., 2020), SiamRCNN (Voigtlaender, Luiten, Torr, & Leibe, 2020), SiamFC++ (Xu et al., 2020), SiamDW (Zhang & Peng, 2019), SiamRPN++ (Li, Wu et al., 2019), SiamRPN (Li, Yan et al., 2018), and SiamFC (Bertinetto et al., 2016), is shown in Fig. 9. Among the compared trackers, SiamEXTR achieved the fourth-best AUC score of 0.697, which was on par with the performance of the third-best tracker, SiamRCNN. It further outperformed the well-known SiamRPN with an AUC scores of 0.637 by a absolute gain of 6.0%. Compared with SiamFC with an AUC score of 0.578, SiamEXTR improved significantly

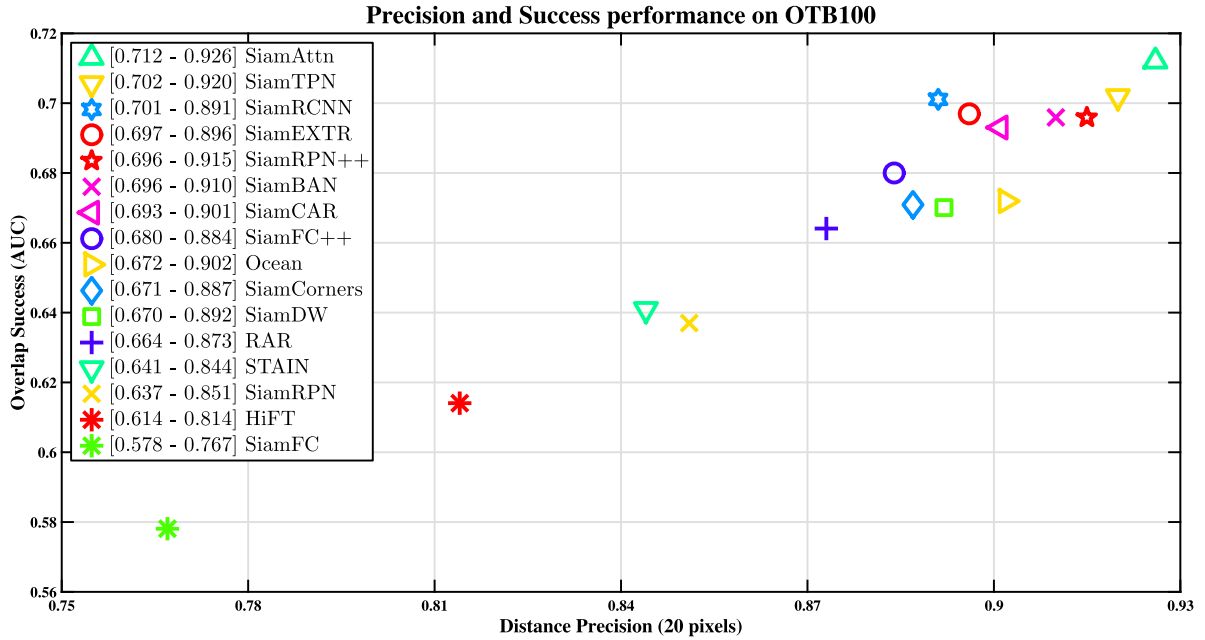


Fig. 9. Precision and success performance on the OTB100 benchmark dataset (Wu et al., 2015). The first number in the legend denotes the AUC score of OSR within the ratio threshold range of [0, 1] while the second indicates the DP score at the error threshold of 20 pixels. The best trackers are located at the top-right corner.

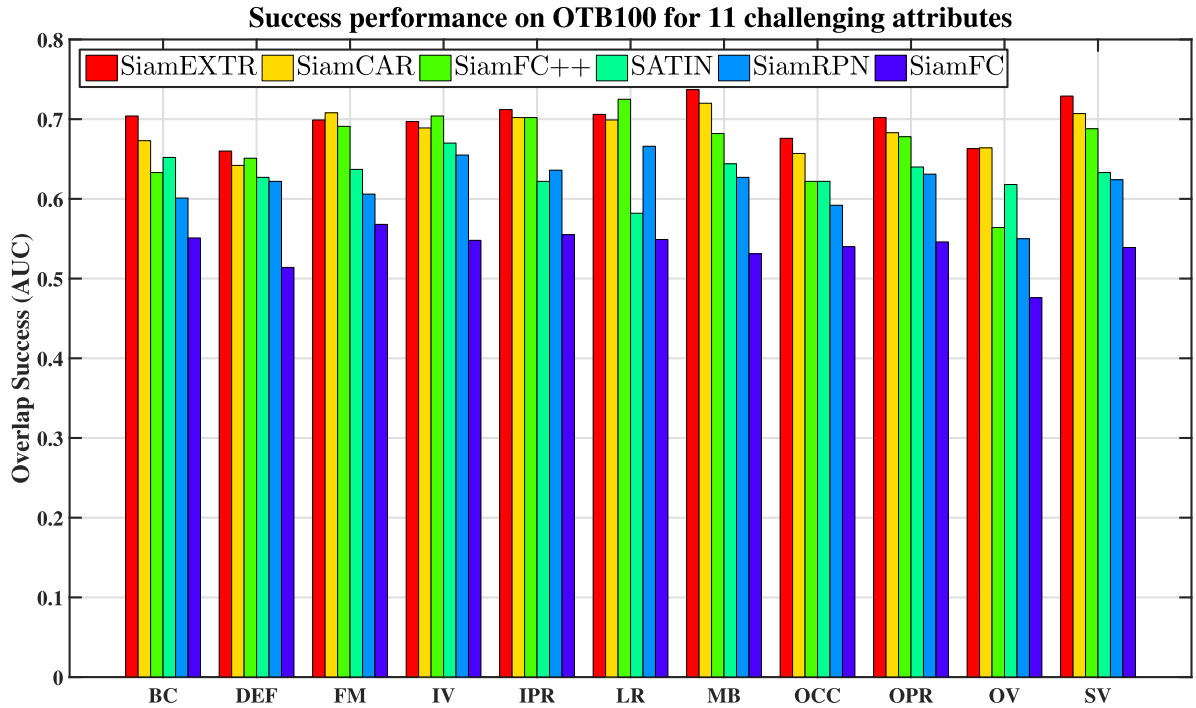


Fig. 10. Success performance of SiamEXTR and other five Siamese trackers over different attributes on the OTB100 benchmark dataset (Wu et al., 2015).

by 11.9%. SiamTPN utilizes a Transformer pyramid network (TPN) to fuse the multilevel convolutional features and obtains an AUC score of 0.702. Without using an extra attention mechanism, SiamEXTR was negligibly inferior to SiamTPN, with an unobtrusive decrease of 0.5% in the AUC score. SiamAttn exploits a deformable attention mechanism to learn a more informative representation and achieves the best DP score of 0.926 on OTB100; its AUC score of 0.712 is slightly superior to that of SiamEXTR by an exceedingly small gain of 1.5%. This demonstrates the effectiveness of richer feature representations extracted from the U-shaped backbone architecture and extreme-pooling

operation. Although HiFT employs a Transformer (Vaswani et al., 2017) to capture the spatial cues of the target object at different scales, its performance is still inferior to SiamEXTR, with an AUC score degradation of 15.3%. SiamCorners is a novel anchor-free tracking approach that can directly track a target object as a pair of corners, achieving an AUC score of 0.671. As a new anchor-free baseline, SiamEXTR outperforms SiamCorners with an absolute gain of 2.8%, demonstrating the advantage of using an extreme point graph for robust annotation. Moreover, SiamEXTR surpasses other up-to-date anchor-free trackers such as SiamCAR, SiamFC++, Ocean, and STAIN, with



Fig. 11. Sample tracking results of the proposed SiamEXTR on eight challenging video sequences *BlurBody*, *BlurOwl*, *carScale*, *coke*, *David3*, *Diving*, *Girl2*, and *tiger1* from the OTB100 benchmark dataset. The ground-truth annotations are illustrated in green.

absolute AUC gains of 0.4%, 1.7%, 2.5%, and 5.6%, respectively. Further, SiamEXTR performed better than the hierarchical anchor-based tracker SiamRPN++ with an absolute AUC gain of 0.1%. In addition, SiamEXTR achieved absolute AUC improvements of 2.7% and 3.3% over the other state-of-the-art SIAMNET-based trackers SiamDW and RAR, respectively. Overall, the comparison on OTB100 benchmark dataset illustrates that the proposed SiamEXTR performs favorably against state-of-the-art tracking approaches.

For a detailed evaluation and analysis of the efficiency and effectiveness of SiamEXTR, it was compared with five representative SIAMNET-based tracking approaches, including three anchor-free trackers, SiamCAR (Cui, Guo et al., 2022), SiamFC++ (Xu et al., 2020), and SATIN (Gao, Yuan et al., 2020), and two baselines, SiamRPN (Li, Yan et al., 2018) and SiamFC (Bertinetto et al., 2016), with different attributes on the OTB100 benchmark dataset. The video sequences in the dataset were annotated with 11 challenging attributes: illumination variation (IV), out-of-plane rotation (OPR), scale variation (SV), occlusion (OCC), deformation (DEF), motion blur (MB), fast motion

(FM), in-plane rotation (IPR), out of view (OV), background clutter (BC) and low resolution (LR). Fig. 10 shows the DP and AUC performances of these trackers with respect to the attributes. SiamEXTR performs favorably against other Siamese trackers for 7 out of the 11 attributes, especially SV, MB, DEF, and BC. These results are partly attributed to the effectiveness of the extreme point graphs detection, which has a much better ability to manage different challenging scenarios during tracking. However, SiamEXTR performed slightly worse than the other trackers in terms of IV, FM, OV, and LR. The primary reason for this is that the extreme point graph-guided annotation utilized in SiamEXTR is sensitive to these scenarios and some target-specific extreme point representations may be lost during feature extraction.

To gain more insight into the performance of the proposed tracker in dealing with these challenging attributes, Fig. 11 shows eight sample tracking results of SiamEXTR on the OTB benchmark dataset, including *BlurBody* (annotated with attributes of SV, DEF, MB, FM, and IPR), *BlurOwl* (annotated with attributes of SV, MB, FM, and IPR), *carScale* (annotated with attributes of SV, OCC, FM, IPR, and OPR),

coke (annotated with attributes of IV, OCC, FM, IPR, OPR, and BC), *David3* (annotated with attributes of OCC, DEF, OPR, and BC), *Diving* (annotated with attributes of SV, DEF, and IPR), *Girl2* (annotated with attributes of SV, OCC, DEF, MB, and OPR), and *tiger1* (annotated with attributes of IV, OCC, DEF, MB, FM, IPR, and OPR). With the primary and refinement networks, the proposed U-shaped backbone can capture sufficient fine-grained informative representations and spatial visual details at the subpixel-level to cope with the substantial shape and appearance alterations caused by BC, FM, OCC, OPR, and OV. Because the sizes and aspect-ratios are determined in a straightforward manner by the detected extreme point graph rather than by preset anchors or scale pyramids, SiamEXTR can significantly mitigate the drastic scale changes caused by DEF, OCC, OPR, and SV, and annotate the target object with a more accurate enclosing bounding box. All the results indicate that the proposed approach can estimate the size and location of the target object better than other complicated heuristic trackers.

4.3. Results on VOT

The performance of SiamEXTR was assessed on VOT2016 (Kristan, Leonardis et al., 2016) and VOT2018 (Kristan et al., 2018) benchmark datasets with respect to A, R, and EAO. Fig. 12 shows the results of SiamEXTR against all participating trackers in VOT2016 and VOT2018 with respect to the EAO score. The EAO scores of the baseline trackers, SiamRPN and SiamFC, on VOT2016 were 0.344 and 0.235, respectively, whereas those on VOT2018 were 0.294 and 0.188, respectively. The proposed tracker obtained the third-best EAO score of 0.497 on VOT2016 and the fourth-best EAO score of 0.428 on VOT2018, significantly outperforming these two trackers by a large margin. Specifically, SiamEXTR surpassed CCOT (Danelljan et al., 2016), the winner of the VOT2016 competition, and LADCF (Kristan et al., 2018), the winner of the VOT2018 competition, with impressive absolute EAO gains of 14.6% and 3.9%, respectively.

Further detailed results are summarized in Table 3 with respect to the EAO, A, R, and speed of 15 up-to-date trackers, including ATOM (Danelljan et al., 2019), CCOT (Danelljan et al., 2016), DaSiamRPN (Zhu, Wang et al., 2018), ECO (Danelljan, Bhat et al., 2017), Ocean (Zhang et al., 2020), SiamAttn (Yu et al., 2020), SiamBAN (Chen et al., 2020), SiamFC (Bertinetto et al., 2016), SiamFC++ (Xu et al., 2020), SiamRCNN (Voigtlaender et al., 2020), SiamRPN (Li, Yan et al., 2018), SiamRPN++ (Li, Wu et al., 2019), and the proposed SiamEXTR. SiamAttn and Ocean achieved the best EAO scores of 0.537 and 0.489 on VOT2016 and VOT2018, respectively, which were superior to those of any other participating tracker in the corresponding competitions. In terms of robustness, the best performance on the two competitions is shown by SiamAtt (0.233) and Ocean (0.117), whereas SiamEXTR attains the third-best scores of 0.153 and 0.166, which are at par with the second most robust trackers SiamBAN (0.150 on VOT2016) and SiamAttn (0.160 on VOT2018), respectively. This can be derived primarily from the use of an extreme-pooling operation to enhance the representation capability for extreme point graph detection. Most remarkably, SiamAttn, SiamBAN, and Ocean achieved the top three EAO scores on VOT2018; however, their speeds were significantly inferior to SiamEXTR. These results demonstrate that SiamEXTR can achieve a more harmonious tracking performance in terms of accuracy, robustness, and speed.

4.4. Results on UAV123

The UAV123 benchmark dataset (Mueller et al., 2016) is always used to evaluate the tracking approaches for UAV applications. The proposed SiamEXTR was compared with 15 well-known trackers, i.e., CLNet (Dong, Shen, Porikli, Luo, & Shao, 2022), SiamCAR (Cui, Guo et al., 2022), SiamTPN (Xing et al., 2022), HiFT (Cao et al., 2021), SiamCorners (Yang et al., 2021), Ocean (Zhang et al., 2020), RAR (Gao, Zhang et al., 2020), SATIN (Gao, Yuan et al., 2020), SiamAttn (Yu et al.,

2020), SiamBAN (Chen et al., 2020), SiamRCNN (Voigtlaender et al., 2020), SiamDW (Zhang & Peng, 2019), SiamRPN++ (Li, Wu et al., 2019), SiamRPN (Li, Yan et al., 2018), and SiamFC (Bertinetto et al., 2016). The DP and AUC of the OSR were adopted for the evaluation.

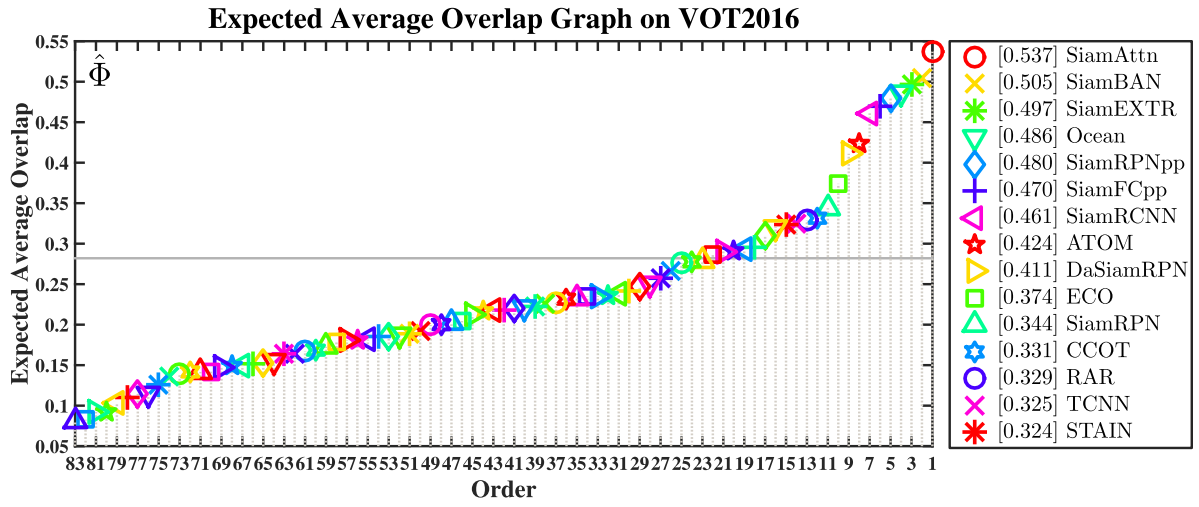
As shown in Fig. 13, SiamEXTR achieved comparable performance in terms of both AUC and DP, with scores of 0.638 and 0.822, respectively. SiamAttn and SiamRCNN attained the top two scores in terms of AUC, outperforming SiamEXTR with slight gaps. SiamEXTR achieved a faster running speed around 142 fps compared to 41 fps for SiamAttn and 22 fps for SiamRCNN. Therefore, the proposed tracker will be improved further by exploiting the attentional mechanism and online learning strategy to narrow these gaps and maintain similar running speeds. In addition, the performance of SiamEXTR is similar to that of the Transformer-based tracker, SiamTPN. This is attributed to the use of extreme-pooling operations, which may drive the backbone network to learn attentional information regarding the target object. The proposed tracker outperformed other anchor-free approaches, including SiamCAR, SiamCorners, and STAIN, in both metrics, with significant relative gains of 1.5%, 2.4%, and 8.6% in terms of AUC, and 0.9%, 0.3%, and 3.8% in terms of DP, respectively. Because the viewpoint of UAV123 is a UAV and the target objects to be tracked are generally small, the performance promotion was attributed to the use of the U-shaped backbone network, which significantly enhances the representation capability for smaller target objects.

All video sequences in the UAV123 benchmark dataset are annotated with several different attributes such as aspect ratio change (ARC), BC, camera motion (CM), FM, full occlusion (FO), IV, LR, OV, partial occlusion (PO), similar objects (SO), SV, and viewpoint change (VC). SiamEXTR achieved a competitive performance compared to other SiamNET-based trackers, including SiamTPN (Xing et al., 2022), SiamDW (Zhang & Peng, 2019), HiFT (Cao et al., 2021), Ocean (Zhang et al., 2020), and SiamBAN (Chen et al., 2020), for every attribute on the UAV123 benchmark dataset, as shown in Fig. 14. SiamEXTR outperformed the other participants in 9 of the 12 challenging attributes, especially in obtaining appealing improvements compared with the Transformer-based tracker, SiamTPN. BC, FO, LR, PO, and SV are the most essential attributes in the UAV123 benchmark dataset, and SiamEXTR successfully managed these scenarios because of the guidance provided by the extreme point graph. However, the performance of SiamEXTR in the OV scenario is slightly lower than that of SiamTPN, Ocean, and SiamBAN because the extreme-pooling operation cannot sufficiently capture the target-specific representation when the target object disappears from the viewpoint. It is noteworthy that all trackers performed poorly in terms of the FO attributes, which implies that research on high performance UAV visual tracking systems must be continuously and intensively conducted.

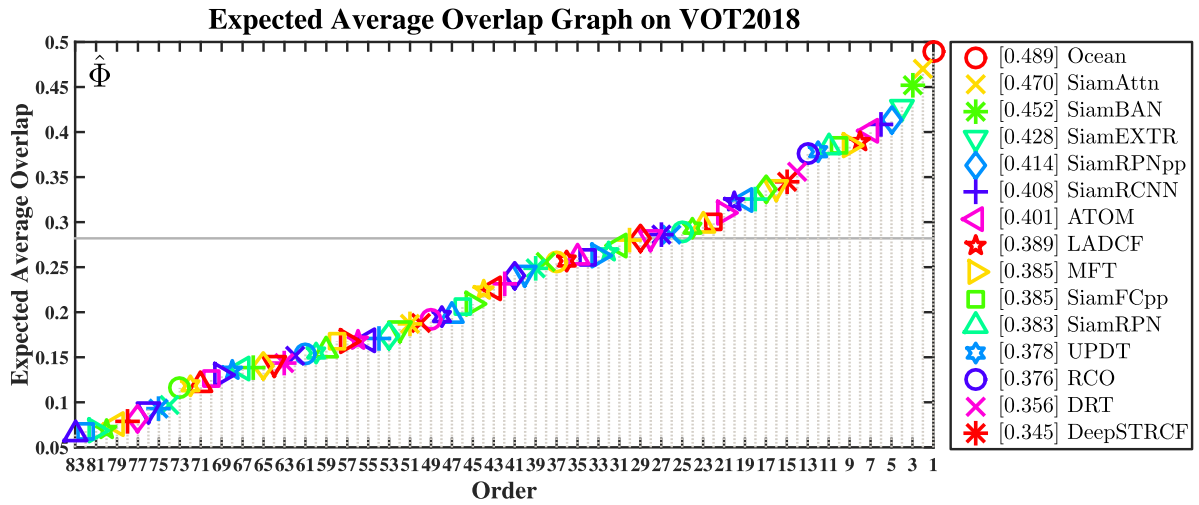
4.5. Results on LaSOT

To further illustrate the effectiveness of the proposed approach, SiamEXTR was compared with top-tier trackers and baselines including CLNet (Dong et al., 2022), SiamCAR (Cui, Guo et al., 2022), SiamTPN (Xing et al., 2022), HiFT (Cao et al., 2021), Ocean (Zhang et al., 2020), SiamBAN (Chen et al., 2020), SiamFC++ (Xu et al., 2020), SiamDW (Zhang & Peng, 2019), SiamRPN++ (Li, Wu et al., 2019), and SiamFC (Bertinetto et al., 2016) on the LaSOT (Fan et al., 2021) benchmark dataset. Three evaluation metrics were used to report the comparison results, including the AUC of OSR, DP, and DP_{Norm} , as shown in Table 4.

SiamEXTR attained favorable performance with the second-best AUC score of 0.532, the second-best DP score of 0.537, and a DP_{Norm} score of 0.582. SiamTPN achieved the best performance on the LaSOT benchmark dataset in terms of AUC, DP, and DP_{Norm} as it trains the Transformer architecture to capture strong discriminative spatial cues of the target object. Compared with the first-rate anchor-free tracker Ocean, SiamEXTR improved the score by an absolute gain of 1.5% in



(a) Expected average overlap performance on VOT2016 (Kristan et al., 2016a).



(b) Expected average overlap performance on VOT2018 (Kristan et al., 2018).

Fig. 12. Expected average overlap performance on VOT challenge datasets. The horizontal gray lines in each plot are the state-of-the-art bounds proposed by the VOT Challenges Committee. For presentation clarity, only the top 15 trackers with respect to the EAO score are displayed in the legend. The better trackers are located at the right.

Table 3

Detailed comparisons with state-of-the-art trackers in terms of EAO, Accuracy and Robustness on the VOT2016 (Kristan, Leonardi et al., 2016) and VOT2018 (Kristan et al., 2018) benchmark datasets. The first, second and third best values are highlighted in red, blue and green fonts, respectively.

Tracker	VOT2016			VOT2018			Speed (fps)
	EAO ↑	A ↑	R ↓	EAO ↑	A ↑	R ↓	
ATOM	0.424	0.617	0.190	0.401	0.590	0.204	30
CCOT	0.331	0.539	0.238	0.267	0.494	0.318	1
DaSiamRPN	0.411	0.610	0.220	0.326	0.570	0.337	164
ECO	0.374	0.546	0.202	0.280	0.484	0.276	8
Ocean	0.486	0.630	0.160	0.489	0.592	0.117	25
SiamAttn	0.537	0.680	0.140	0.470	0.630	0.160	41
SiamBAN	0.505	0.632	0.150	0.452	0.597	0.178	52
SiamFC	0.235	0.532	0.461	0.188	0.503	0.585	90
SiamFC++	0.470	0.624	0.154	0.385	0.577	0.183	75
SiamRCNN	0.461	0.645	0.173	0.408	0.609	0.220	22
SiamRPN	0.344	0.560	0.261	0.383	0.586	0.276	170
SiamRPN++	0.480	0.580	0.197	0.414	0.600	0.234	82
SiamEXTR	0.497	0.637	0.153	0.428	0.582	0.166	142

AUC, 0.9% in DP, and 2.1% in DP_{Norm}, respectively. This is chiefly attributed to the accurate guidance provided by the extreme point graph for annotation. Another promising anchor-free tracker, SiamCAR, achieved the second-best DP_{Norm} score of 0.600; however, its AUC

and DP scores were significantly inferior to those of SiamEXTR, with severe drops of 2.5% and 2.7%, respectively. The experimental results on such a large-scale benchmark dataset demonstrate the superiority of the proposed SiamEXTR compared with up-to-date excellent trackers.

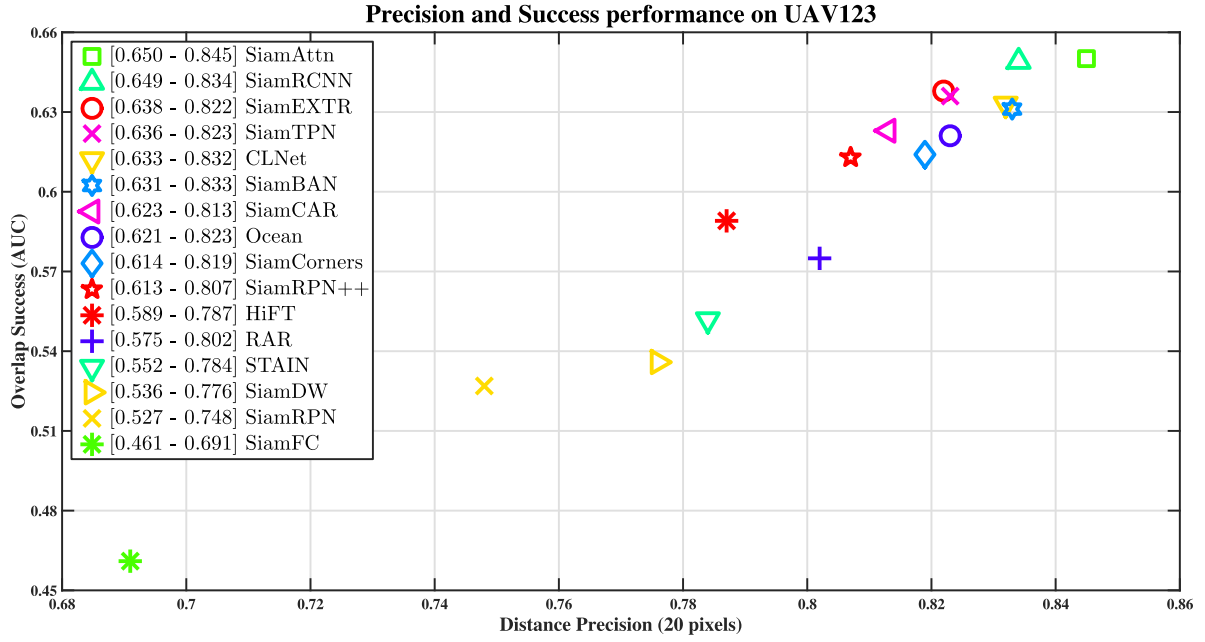


Fig. 13. Precision and success performance on the UAV123 benchmark dataset (Mueller et al., 2016). The first number in the legend denotes AUC score of OSR within the ratio threshold range of [0, 1] while the second indicates the DP score at the error threshold of 20 pixels. The best trackers are located at the top-right corner.

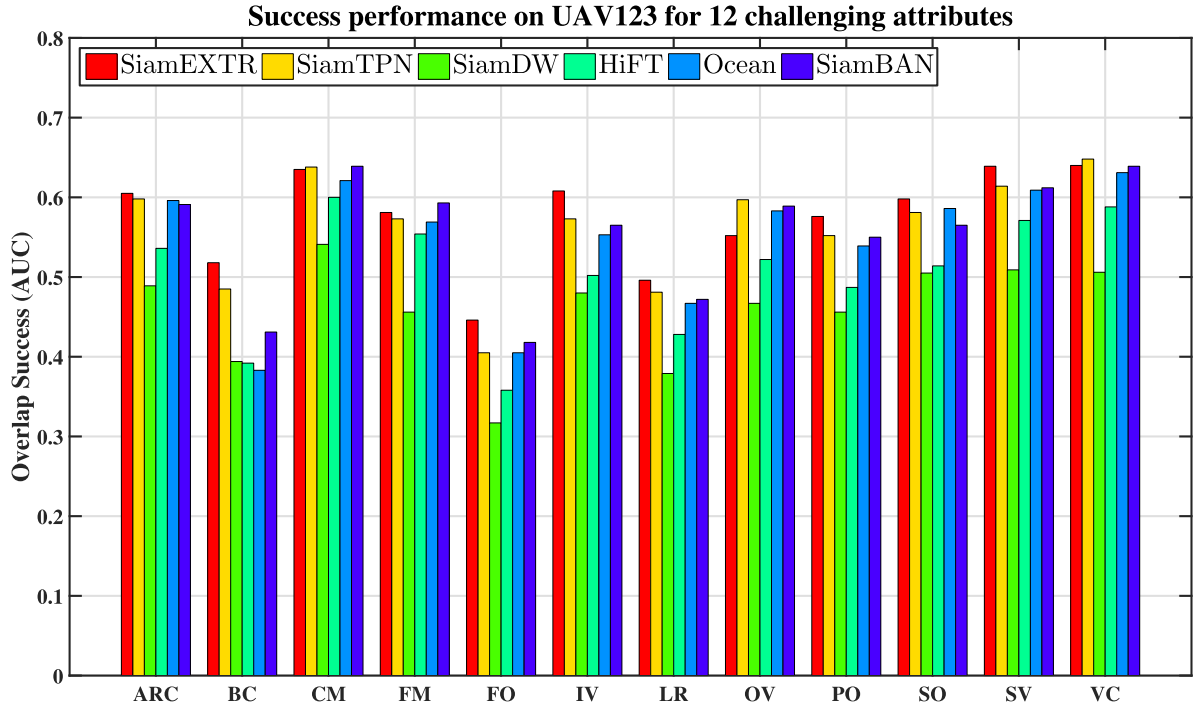


Fig. 14. Success performance of SiamEXTR and other five excellent trackers over different attributes on the UAV123 benchmark dataset (Mueller et al., 2016).

The target objects in the LaSOT benchmark dataset are fully annotated with detailed attributes, including ARC, BC, CM, DEF, FM, FO, IV, LR, MB, OV, PO, rotation (ROT), SV, and VC. Based on the attribute-wise comparison shown in Fig. 15, the Transformer-based tracker SiamTPN achieved the best performance in the most challenging scenarios. It is believed that this is primarily because the Transformer architecture employed in SiamTPN is well suited for extracting temporal cues from long-term video sequences. Although SiamEXTR is not as exceptional as SiamTPN in the LaSOT benchmark dataset, it still

outperforms the latter by a large margin in the PO scenarios because it can quickly detect the four extreme point positions of the target object after occlusion. Among these competitors, SiamFC++ significantly outperformed SiamEXTR in the ARC, DEF, and VC scenarios, which was attributed to the advantages of the decomposition strategy and estimation quality assessment employed by SiamFC++. The experimental results clearly demonstrate that the attention mechanism plays a crucial role in achieving robust visual tracking in long-term video sequences.

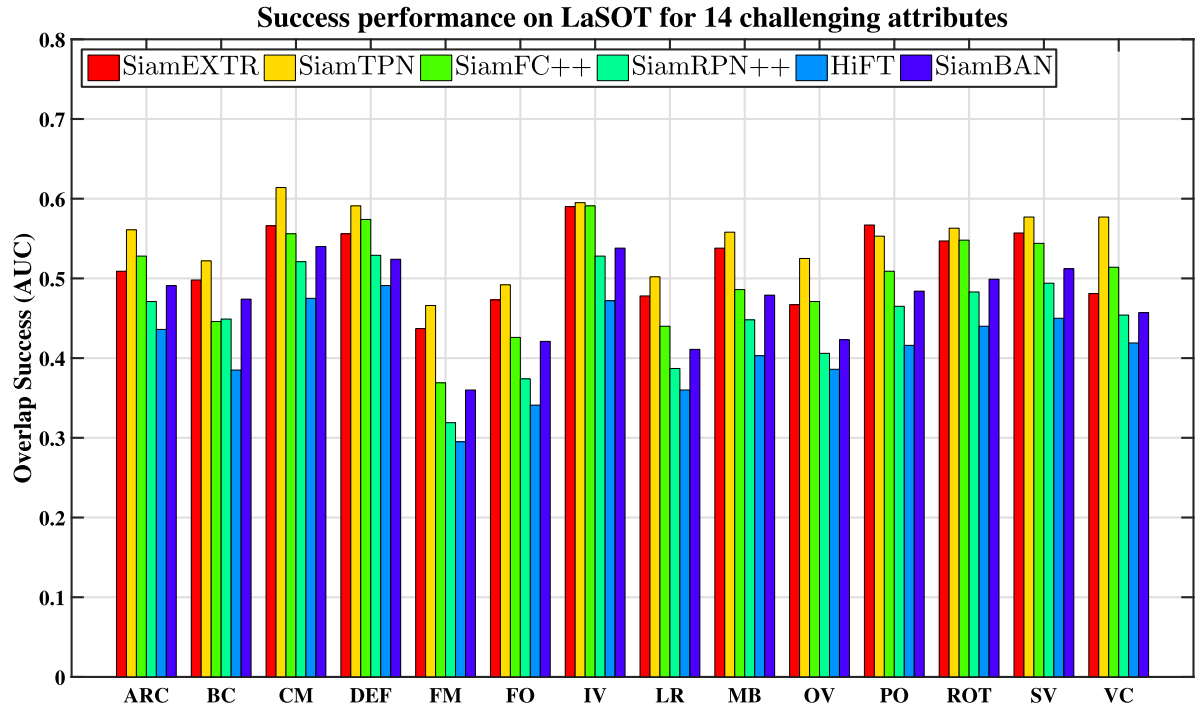


Fig. 15. Success performance of SiamEXTR and other five excellent trackers over different attributes on the LaSOT benchmark dataset (Fan et al., 2021).

Table 4

Comparison results of top-tier trackers in terms of success rate, precision and normalized precision on the LaSOT benchmark dataset (Fan et al., 2021). The first, second and third best values are highlighted in red, blue and green fonts, respectively.

Tracker	AUC	DP	DP _{Norm}
CLNet	0.499	0.494	0.574
HiFT	0.451	0.421	0.527
Ocean	0.517	0.526	0.561
SiamBAN	0.514	0.521	0.598
SiamCAR	0.507	0.510	0.600
SiamDW	0.385	0.389	0.480
SiamFC	0.336	0.339	0.420
SiamFC++	0.501	0.475	0.525
SiamRPN++	0.496	0.491	0.569
SiamTPN	0.581	0.578	0.683
SiamEXTR	0.532	0.537	0.582

Table 5

Detailed comparisons with state-of-the-art trackers in terms of average overlap and success rate on the GOT-10k benchmark dataset (Huang et al., 2019). The first, second and third best values are highlighted in red, blue and green fonts, respectively.

Tracker	AO	SR _{0.50}	SR _{0.75}
ATOM	0.556	0.634	0.402
Ocean	0.611	0.721	0.473
SiamCAR	0.569	0.670	0.415
SiamDW	0.429	0.483	0.147
SiamFC	0.392	0.426	0.135
SiamFC++	0.595	0.695	0.479
SiamMask	0.453	0.550	0.248
SiamRPN	0.481	0.581	0.270
SiamRPN++	0.518	0.618	0.329
SiamTPN	0.576	0.686	0.441
SiamEXTR	0.595	0.701	0.462

4.6. Results on GOT-10k

Table 5 shows the comparison results of SiamEXTR against 10 state-of-the-art trackers on GOT-10k (Huang et al., 2019), including SiamTPN (Xing et al., 2022), SiamCAR (Cui, Guo et al., 2022), Ocean (Zhang et al., 2020), SiamFC++ (Xu et al., 2020), ATOM (Danelljan et al., 2019), SiamRPN++ (Li, Wu et al., 2019), SiamMask (Wang, Zhang et al., 2019), SiamDW (Zhang & Peng, 2019), SiamRPN (Li, Yan et al., 2018), and SiamFC (Bertinetto et al., 2016). The official evaluation metrics AO, SR_{0.50} and SR_{0.75} were employed as the performance indicators.

SiamEXTR obtained the second-best AO score of 0.595, the second-best SR_{0.50} score of 0.701, and the third-best SR_{0.75} score of 0.462. Compared to the representative anchor-free tracker SiamCAR, SiamEXTR improved the scores by 2.6%, 3.1%, and 4.7% in terms of AO, SR_{0.50} and SR_{0.75}, respectively, demonstrating the efficiency of the proposed extreme-pooling operation. Another anchor-free tracker SiamFC++ provided the best SR_{0.75} score of 0.479, but its AO and SR_{0.50} scores were inferior to those of SiamEXTR. Although the Ocean tracker achieved the highest AO score of 0.611 and SR_{0.50} score of 0.721, the gap between SiamEXTR and it was small. The SiamTPN tracker exhibited inferior performance on this dataset. Because the GOT-10k benchmark dataset has a large diversity of target object classes and certain classes are not included in the training split, it is believed that one of the reasons for this is that the Transformer architecture employed by SiamTPN is not as effective in handling unfamiliar objects, thus limiting its generalization capability. These results show that SiamEXTR can achieve competitive tracking performance with remarkable generalization capability.

4.7. Discussions

To verify the contributions of the design choices, SiamEXTR was compared with several variants on the OTB100 (Wu et al., 2015) and UAV123 (Mueller et al., 2016) benchmark datasets. For completeness, the baseline Siamese anchor-free tracker, SiamCorners (Yang et al., 2021), was also intentionally added for the evaluation. Table 6 presents the detailed results with respect to the DP and AUC of the OSR.

Table 6

Ablation studies of several variations of SiamEXTR on OTB100 and UAV123 benchmarks using AUC and DP metrics.

Tracker	OTB100		UAV123	
	AUC	DP	AUC	DP
SiamCorners	0.671	0.887	0.614	0.819
SiamEXTR _{ResNet}	0.634	0.852	0.567	0.774
SiamEXTR _{U-Net}	0.653	0.866	0.581	0.779
SiamEXTR _{Primary}	0.675	0.881	0.602	0.788
SiamEXTR _{Corners}	0.684	0.889	0.611	0.798
SiamEXTR _{w/o-pooling}	0.534	0.695	0.467	0.644
SiamEXTR _{w/o-L_{cp}}	0.661	0.866	0.609	0.784
SiamEXTR _{w/o-L_{os}}	0.679	0.885	0.613	0.792
SiamEXTR _{w/o-L_{sym}}	0.654	0.873	0.587	0.785
SiamEXTR	0.697	0.896	0.638	0.822

4.7.1. U-shaped backbone network

The three variants, i.e., SiamEXTR_{ResNet}, SiamEXTR_{U-Net} and SiamEXTR_{Primary} were implemented to compared with different network architectures for deep feature extraction, which employ ResNet-50 (He, Zhang, Ren, & Sun, 2016),¹ U-Net (Ronneberger et al., 2015), and the proposed primary network as backbones. It is worth noting that neither variant exploited the refinement network to further enhance its representation capability. It was empirically observed that replacing ResNet-50 with U-Net steadily improved the AUC score by 1.9%, from 0.634 to 0.653, and the DP score by 1.4%, from 0.852 to 0.866 on OTB100, respectively. The performance can be further improved by utilizing the proposed primary backbone network. The AUC score improved by 2.2%, from 0.653 to 0.675, and the DP score increased by 1.5%, from 0.866 to 0.881. The indicators on UAV123 were further improved by at least 0.9%–2.1%. Jointly employing both primary and refinement networks makes SiamEXTR more accurate and robust. This resulted in a high AUC score of 0.638 on UAV123, surpassing that of SiamCorners by at least 2.4%. This is because the proposed U-shaped backbone network can alleviate the impact of the low-resolution feature maps, which lack important fine-grained visual details for extreme point graph detection, and thus improve the performance.

4.7.2. Guidance of extreme point graph

To demonstrate the advantages of using extreme point graphs for visual tracking, a variant SiamEXTR_{Corners} was implemented that tracks the target object as a pair of opposing corner points (the top-left and bottom-right corner points) of the bounding box (Law & Deng, 2018), similar to SiamCorners. The results showed that SiamEXTR_{Corners} obtained promising performance, outperforming SiamCorners with 1.3% and 0.2% gains in terms of AUC and DP on OTB100, but was inferior to SiamEXTR. This evidence indicates that extreme point graphs are more suitable for visual tracking than opposing corner points.

4.7.3. Extreme-pooling operation

To analyze the effectiveness of the proposed extreme-pooling operation, all oriented max-pooling operations were removed from SiamEXTR and evaluated the variant SiamEXTR_{w/o-pooling} which directly uses the backbone feature maps for extreme point graph detection. Compared with SiamEXTR, the performance of SiamEXTR_{w/o-pooling} decreased by 16.3% and 17.1% in terms of AUC, and by 20.1% and 17.8% in terms of DP on OTB100 and UAV123, respectively. As shown in Fig. 16, with the proposed extreme-pooling operation, the detection heatmaps can focus more accurately on the extreme points of the target object. This suggests that extreme-pooling operations play a significant role in extreme point graph detection.

Table 7

Evaluation results of different annotation strategies on the OTB2015 and VOT2018 benchmark datasets.

Tracker	OTB2015		VOT2018		
	AUC	DP	EAO ↑	A ↑	R ↓
SiamEXTR _{Mask}	0.681	0.896	0.443	0.611	0.166
SiamEXTR	0.697	0.896	0.428	0.582	0.166

4.7.4. Multi-task tracking loss

The multi-task tracking loss is an essential component of the proposed approach. It comprises the extreme points location loss, center point location loss, offset loss, and symmetric loss. The first is the fundamental component of multi-task tracking loss. To evaluate the impact of the last three auxiliary losses, SiamEXTR was compared against three variants SiamEXTR_{w/o-L_{cp}}, SiamEXTR_{w/o-L_{os}} and SiamEXTR_{w/o-L_{sym}}, which were obtained by removing the center point location loss, the offset loss, and the symmetric loss from SiamEXTR, respectively. As presented in Table 6, the AUC and DP scores of SiamEXTR on UAV123 decreased by 3.4% and 3.8%, respectively, without using the center point location loss, and by 2.5% and 3.0%, respectively, without using the offset loss. By removing the symmetric loss, the proposed tracker obtained lower AUC scores, with 4.3% and 5.1% reductions on OTB100 and UAV123, respectively. These performance degradations clearly indicate that multi-task tracking loss contributes to obtaining more accurate tracking results and symmetric loss plays a more crucial role in training SiamEXTR.

4.7.5. Annotation strategy

An additional experiment was conducted on the OTB2015 and VOT2018 benchmark datasets to investigate the effects of using octagonal masks instead of traditional axis-aligned bounding boxes to annotate objects. The results are listed in Table 7. It is observed that SiamEXTR_{Mask} drops the AUC score by 1.6% from 0.697 to 0.681 on OTB2015 compared to SiamEXTR. However, SiamEXTR_{Mask} gains 1.5% improvement in terms of EAO and a 3.8% improvement in terms of accuracy, respectively. This is partly because some ground-truth annotations in VOT2018 are manually rotated, whereas the OTB100 benchmark always uses axis-aligned rectangles to annotate target objects.

4.8. More discussions

Over the last three years, Transformer-based trackers (Thangavel et al., 2023) have shown excellent performance compared to non-Transformer-based trackers and are no surprise dominating the visual tracking community. SiamEXTR was compared with five more recent Transformer-based trackers: DropMAE (Wu et al., 2023), SeqTrack (Chen et al., 2023), MixFormer (Cui, Jiang et al., 2022), OS-Track (Ye et al., 2022), and SwinTrack (Lin, Fan, Zhang, Xu, & Ling, 2022), on the LaSOT and GOT-10k benchmark datasets.

As shown in Table 8, the Transformer-based trackers outperformed SiamEXTR by a wide margin in all experimental metrics because they inherit the global feature learning capabilities of the Transformer architectures. In addition, the computational efficiency of these tracker was demonstrated in terms of the FLOPs and speed (fps). It is evident that all the Transformer-based trackers exhibit poor efficiency compared to SiamEXTR. It is believed that the complex attention mechanism employed by the Transformer architecture results in an increase in the quadratic computational cost. Although, SiamEXTR successfully achieved a considerable balance between the tracking robustness and efficiency in benchmark datasets by treating visual tracking as an extreme point graph detection task, further attempts will be made to design a backbone network based on the Transformer architecture to enhance target-specific discriminative features and eliminate background and distractor features.

¹ The strides of last three stages *conv3*, *conv4* and *conv5* are set to 8.



Fig. 16. Visualization of the detection heatmaps of extreme point graph. (a) is the searching candidate with ground-truth annotation. (b) and (c) are the visualized images by summing four extreme point detection heatmaps generated with and without the extreme-pooling operation, respectively.

Table 8

Comparison of SiamEXTR with recent Transformer-based trackers on the LaSOT benchmark dataset in terms of AUC, DP and DP_{Norm} scores, and on the GOT-10k benchmark dataset in terms of mAO, $mSR_{0.50}$ and $mSR_{0.75}$ scores.

Tracker	LaSOT			GOT-10k			FLOPs (G)	Speed (fps)
	AUC	DP	DP_{Norm}	AO	$SR_{0.50}$	$SR_{0.75}$		
DropMAE	0.715	0.779	0.815	0.759	0.868	0.720	48.32	16
SeqTrack	0.725	0.792	0.815	0.748	0.819	0.722	535.85	4
MixFormer	0.700	0.763	0.799	0.707	0.800	0.678	113.02	16
OSTrack	0.711	0.776	0.811	0.737	0.832	0.708	48.32	16
SwinTrack	0.702	0.753	0.784	0.694	0.780	0.643	61.85	9
SiamEXTR	0.532	0.537	0.582	0.595	0.701	0.462	8.85	142

5. Conclusion

In this study, SiamEXTR, a novel fully-convolutional approach that formulates generic visual tracking as an extreme point graph detection problem is proposed to improve both tracking robustness and accuracy. The proposed approach is based on three main principles: (i) a series of extreme-pooling operations that convert the bounding box estimation of the target object into an extreme point graph detection method, (ii) a U-shaped backbone network that captures fine-grained visual representation and strong semantic information at high-resolution to guarantee accurate extreme point graph detection, and (iii) a multi-task tracking loss that ensures robustness in the extreme point graph detection process. As a result, based on the detected extreme point graph, SiamEXTR not only produces axis-aligned bounding boxes to annotate target objects but also obtains more desirable octagonal segmentation masks through a simple approximate strategy. To the best of our knowledge, this is the first study to annotate target objects under the guidance of an extreme point graph in a unified SIAMNET framework without foreground-background classification or region proposal regression. Without bells and whistles, extensive quantitative and qualitative experiments on six popular dense benchmark datasets—OTB100,

VOT2016, VOT2018, UAV123, LaSOT, and GOT-10k—clearly demonstrated that SiamEXTR consistently achieves competitive performance in terms of accuracy and robustness when the running speed far exceeds 140 fps. The proposed SiamEXTR is anticipated to inspire further research to promote the development of the visual tracking community and even the object pose estimation research field (Majcher & Kwolek, 2022; Sun, Liu, Sun, & Park, 2021). As recent visual tracking community have been ignited by Transformer-based approaches (Chen et al., 2021; Yan, Peng, Fu, Wang, & Lu, 2021), future studies will include the incorporation of efficient Transformer architectures into SiamEXTR to provide a more robust performance.

CRedit authorship contribution statement

Peng Gao: Writing – original draft, Writing – review & editing, Conceptualization, Methodology, Investigation, Software. **Xin-Yue Zhang:** Writing – original draft, Writing – review & editing, Visualization, Software. **Xiao-Li Yang:** Conception, Design, Resources. **Feng Gao:** Software, Data curation. **Hamido Fujita:** Writing – original draft, Supervision, Validation. **Fei Wang:** Writing – review & editing, Formal analysis, Validation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Peng Gao reports financial support was provided by China Postdoctoral Science Foundation. Peng Gao reports financial support was provided by Qufu Normal University. Peng Gao reports financial support was provided by Shandong Provincial Natural Science Foundation. Peng Gao reports financial support was provided by Guangdong Provincial Basic and Applied Basic Research Foundation.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported in part by the China Postdoctoral Science Foundation, China under Grant 2023M732022, in part by the Qufu Normal University under Grant 167-602801, in part by the Shandong Provincial Natural Science Foundation under Grant ZR2021QF061, in part by the Guangdong Provincial Basic and Applied Basic Research Foundation under Grant 2020A1515010706, and in part by the sponsorship from gpushare.com. The authors would like to thank all the anonymous reviewers for their insightful comments and suggestions that have helped to significantly improve the quality of this paper.

References

- Badrinarayanan, V., Kendall, A., & Cipolla, R. (2017). Segnet: A deep convolutional encoder-decoder architecture for image segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 39(12), 2481–2495.
- Bertinetto, L., Valmadre, J., Henriques, J., Vedaldi, A., & Torr, P. H. S. (2016). Fully-convolutional siamese networks for object tracking. In *European conference on computer vision (ECCV)* (pp. 850–865). Springer-Verlag.
- Bromley, J., Guyon, I., LeCun, Y., Säckinger, E., & Shah, R. (1993). Signature verification using a “siamese” time delay neural network. In *Advances in neural information processing systems (NeurIPS)*, Vol. 6.
- Cao, Z., Fu, C., Ye, J., Li, B., & Li, Y. (2021). HiFT: Hierarchical feature transformer for aerial tracking. In *International conference on computer vision (ICCV)* (pp. 15457–15466).
- Čehovin, L., Leonardis, A., & Kristan, M. (2016). Visual object tracking performance measures revisited. *IEEE Transactions on Image Processing*, 25(3), 1261–1274.
- Chen, X., Peng, H., Wang, D., Lu, H., & Hu, H. (2023). SeqTrack: Sequence to sequence learning for visual object tracking. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 14572–14581).
- Chen, X., Yan, B., Zhu, J., Wang, D., Yang, X., & Lu, H. (2021). Transformer tracking. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 8126–8135). IEEE.
- Chen, Z., Zhong, B., Li, G., Zhang, S., & Ji, R. (2020). Siamese box adaptive network for visual tracking. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 6668–6677). IEEE.
- Cui, Y., Guo, D., Shao, Y., Wang, Z., Shen, C., Zhang, L., et al. (2022). Joint classification and regression for visual tracking with fully convolutional siamese networks. *International Journal of Computer Vision*, 130(2), 550–566.
- Cui, Y., Jiang, C., Wang, L., & Wu, G. (2022). Mixformer: End-to-end tracking with iterative mixed attention. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 13608–13618).
- Danelljan, M., Bhat, G., Khan, S. F., & Felsberg, M. (2017). ECO: Efficient convolution operators for tracking. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 6931–6939). IEEE.
- Danelljan, M., Bhat, G., Khan, F. S., & Felsberg, M. (2019). Atom: Accurate tracking by overlap maximization. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 4660–4669). IEEE.
- Danelljan, M., Häger, G., Khan, F. S., & Felsberg, M. (2015). Convolutional features for correlation filter based visual tracking. In *International conference on computer vision (ICCV)* (pp. 621–629). IEEE.
- Danelljan, M., Häger, G., Khan, F. S., & Felsberg, M. (2017). Discriminative scale space tracking. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 39(8), 1561–1575.
- Danelljan, M., Robinson, A., Khan, F. S., & Felsberg, M. (2016). Beyond correlation filters: Learning continuous convolution operators for visual tracking. In *European conference on computer vision (ECCV)* (pp. 472–488). Springer-Verlag.
- Dong, X., Shen, J., Porikli, F., Luo, J., & Shao, L. (2022). Adaptive siamese tracking with a compact latent network. *IEEE*.
- Duan, K., Bai, S., Xie, L., Qi, H., Huang, Q., & Tian, Q. (2019). Centernet: Keypoint triplets for object detection. In *International conference on computer vision (ICCV)* (pp. 6569–6578). IEEE.
- Everingham, M., Eslami, S., Van Gool, L., Williams, C. K., Winn, J., & Zisserman, A. (2015). The pascal visual object classes challenge: A retrospective. *International Journal of Computer Vision*, 111(1), 98–136.
- Fan, H., Bai, H., Lin, L., Yang, F., Chu, P., Deng, G., et al. (2021). Lasot: A high-quality large-scale single object tracking benchmark. *International Journal of Computer Vision*, 129(2), 439–461.
- Fan, H., & Ling, H. (2019). Siamese cascaded region proposal networks for real-time visual tracking. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 7952–7961). IEEE.
- Felzenszwalb, P. F., Girshick, R. B., McAllester, D., & Ramanan, D. (2010). Object detection with discriminatively trained part-based models. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 32(9), 1627–1645.
- Fieraru, M., Khoreva, A., Pishchulin, L., & Schiele, B. (2018). Learning to refine human pose estimation. In *IEEE conference on computer vision and pattern recognition workshops (CVPRW)* (pp. 318–327). IEEE.
- Gao, P., Ma, Y., Li, C., Song, K., Zhang, Y., Wang, F., et al. (2018). Adaptive object tracking with complementary models. *IEEE Transactions on Information and Systems*, E101-D(11), 2849–2854.
- Gao, P., Ma, Y., Song, K., Li, C., Wang, F., Xiao, L., et al. (2018). High performance visual tracking with circular and structural operators. *Knowledge-Based Systems*, 161, 240–253.
- Gao, P., Ma, Y., Yuan, R., Xiao, L., & Wang, F. (2019). Learning cascaded siamese networks for high performance visual tracking. In *International conference on image processing (ICIP)* (pp. 3078–3082). IEEE.
- Gao, P., Yuan, R., Wang, F., Xiao, L., Fujita, H., & Zhang, Y. (2020). Siamese attentional keypoint network for high performance visual tracking. *Knowledge-Based Systems*, 193(105448), 1–16.
- Gao, P., Zhang, Q., Wang, F., Xiao, L., Fujita, H., & Zhang, Y. (2020). Learning reinforced attentional representation for end-to-end visual tracking. *Information Sciences*, 517, 52–67.
- Girshick, R. (2015). Fast r-cnn. In *International conference on computer vision (ICCV)* (pp. 1440–1448).
- Girshick, R., Donahue, J., Darrell, T., & Malik, J. (2014). Rich feature hierarchies for accurate object detection and semantic segmentation. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 580–587). IEEE.
- Guo, D., Shao, Y., Cui, Y., Wang, Z., Zhang, L., & Shen, C. (2021). Graph attention tracking. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 9538–9547). IEEE.
- He, A., Luo, C., Tian, X., & Zeng, W. (2018). A twofold siamese network for real-time object tracking. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 4834–4843). IEEE.
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 770–778). IEEE.
- Huang, L., Zhao, X., & Huang, K. (2019). Got-10k: A large high-diversity benchmark for generic object tracking in the wild. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(5), 1562–1577.
- Javed, S., Danelljan, M., Khan, F. S., Khan, M. H., Felsberg, M., & Matas, J. (2022). Visual object tracking with discriminative filters and siamese networks: a survey and outlook. *IEEE Transactions on Pattern Analysis and Machine Intelligence*.
- Kristan, M., Leonardis, A., Matas, J., Felsberg, M., Pflugfelder, R., Čehovin, L., et al. (2016). The visual object tracking VOT2016 challenge results. In *European conference on computer vision (ECCV)*. Springer-Verlag.
- Kristan, M., Leonardis, A., Matas, J., Felsberg, M., Pflugfelder, R., Kämäräinen, J.-K., et al. (2020). The eighth visual object tracking VOT2020 challenge results. In *European conference on computer vision (ECCV)* (pp. 547–601). Springer.
- Kristan, M., Leonardis, A., Matas, J., Felsberg, M., Pflugfelder, R., Čehovin, L., et al. (2018). The sixth visual object tracking VOT2018 challenge results. In *European conference on computer vision (ECCV)*. Springer-Verlag.
- Kristan, M., Matas, J., Leonardis, A., Felsberg, M., Pflugfelder, R., Kamarainen, J.-K., et al. (2019). The seventh visual object tracking vot2019 challenge results. In *International conference on computer vision (ICCV)*. IEEE.
- Kristan, M., Matas, J., Leonardis, A., Vojit, T., Pflugfelder, R., Fernández, G., et al. (2016). A novel performance evaluation methodology for single-target trackers. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 38(11), 2137–2155.
- Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). ImageNet classification with deep convolutional neural networks. In *Advances in neural information processing systems (NeurIPS)* (pp. 1097–1105).
- Law, H., & Deng, J. (2018). Cornernet: Detecting objects as paired keypoints. In *European conference on computer vision (ECCV)* (pp. 734–750).
- LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, 521(7553), 436–444.
- Li, P., Chen, B., Ouyang, W., Wang, D., Yang, X., & Lu, H. (2019). Gradnet: Gradient-guided network for visual object tracking. In *International conference on computer vision (ICCV)* (pp. 6162–6171). IEEE.
- Li, Y., Chen, Y., Wang, N., & Zhang, Z. (2019). Scale-aware trident networks for object detection. *arXiv preprint arXiv:1901.01892*.

- Li, X., Ma, C., Wu, B., He, Z., & Yang, M.-H. (2019). Target-aware deep tracking. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 1369–1378). IEEE.
- Li, P., Wang, D., Wang, L., & Lu, H. (2018). Deep visual tracking: Review and experimental comparison. *Pattern Recognition*, 76, 323–338.
- Li, B., Wu, W., Wang, Q., Zhang, F., Xing, J., & Yan, J. (2019). Siamrpn++: Evolution of siamese visual tracking with very deep networks. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 4282–4291). IEEE.
- Li, H., Xiong, P., An, J., & Wang, L. (2018). Pyramid attention network for semantic segmentation. In *British machine vision conference (BMVC)*. BMVA.
- Li, B., Yan, J., Wu, W., Zhu, Z., & Hu, X. (2018). High performance visual tracking With Siamese Region proposal network. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 8971–8980). IEEE.
- Li, Y., & Zhu, J. (2014). A scale adaptive kernel correlation filter tracker with feature integration. In *European conference on computer vision (ECCV)* (pp. 254–265). Springer-Verlag.
- Lin, T.-Y., Dollár, P., Girshick, R., He, K., Hariharan, B., & Belongie, S. (2017). Feature pyramid networks for object detection. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 2117–2125). IEEE.
- Lin, L., Fan, H., Zhang, Z., Xu, Y., & Ling, H. (2022). Swintrack: A simple and strong baseline for transformer tracking. *Advances in Neural Information Processing Systems*, 35, 16743–16754.
- Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., et al. (2014). Microsoft coco: Common objects in context. In *European conference on computer vision (ECCV)* (pp. 740–755). Springer-Verlag.
- Ma, C., Huang, J.-B., Yang, X., & Yang, M.-H. (2019). Robust visual tracking via hierarchical convolutional features. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 41(11), 2709–2723.
- Majcher, M., & Kwolek, B. (2022). Pose guided feature learning for 3D object tracking on RGB videos. In *VISIGRAPP (5: VISAPP)* (pp. 574–581).
- Maninis, K.-K., Caelles, S., Pont-Tuset, J., & Van Gool, L. (2018). Deep extreme cut: From extreme points to object segmentation. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 616–625). IEEE.
- Marvasti-Zadeh, S. M., Cheng, L., Ghanei-Yakhdan, H., & Kasaei, S. (2022). Deep learning for visual tracking: A comprehensive survey. *IEEE Transactions on Intelligent Transportation Systems*, 23(5), 3943–3968.
- Mueller, M., Smith, N., & Ghanem, B. (2016). A benchmark and simulator for uav tracking. In *European conference on computer vision (ECCV)* (pp. 445–461). Springer.
- Newell, A., Yang, K., & Deng, J. (2016). Stacked hourglass networks for human pose estimation. In *European conference on computer vision (ECCV)* (pp. 483–499). Springer-Verlag.
- Ondrašovič, M., & Tarábek, P. (2021). Siamese visual object tracking: A survey. *IEEE Access*, 9, 110149–110172.
- Peng, J., Jiang, Z., Gu, Y., Wu, Y., Wang, Y., Tai, Y., et al. (2021). Siamrcr: Reciprocal classification and regression for visual object tracking. *arXiv preprint arXiv:2105.11237*.
- Perazzi, F., Pont-Tuset, J., McWilliams, B., Van Gool, L., Gross, M., & Sorkine-Hornung, A. (2016). A benchmark dataset and evaluation methodology for video object segmentation. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 724–732). IEEE.
- Pont-Tuset, J., Perazzi, F., Caelles, S., Arbeláez, P., Sorkine-Hornung, A., & Van Gool, L. (2017). The 2017 davis challenge on video object segmentation. *arXiv preprint arXiv:1704.00675*.
- Qi, Y., Zhang, S., Qin, L., Huang, Q., Yao, H., Lim, J., et al. (2019). Hedging deep features for visual tracking. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 41(5), 1116–1129.
- Ren, S., He, K., Girshick, R., & Sun, J. (2017). Faster r-cnn: Towards real-time object detection with region proposal networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 39(6), 1137–1149.
- Ronneberger, O., Fischer, P., & Brox, T. (2015). U-net: Convolutional networks for biomedical image segmentation. In *International conference on medical image computing and computer-assisted intervention* (pp. 234–241). Springer.
- Russakovsky, O., Deng, J., Su, H., Krause, J., Satheesh, S., Ma, S., et al. (2015). Imagenet large scale visual recognition challenge. *International Journal of Computer Vision*, 115(3), 211–252.
- Schmidhuber, J. (2015). Deep learning in neural networks: An overview. *Neural Networks*, 61, 85–117.
- Simonyan, K., & Zisserman, A. (2015). Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556v6*.
- Smeulders, A. W., Chu, D. M., Cucchiara, R., Calderara, S., Dehghan, A., & Shah, M. (2014). Visual tracking: An experimental survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 36(7), 1442–1468.
- Sun, S., Liu, R., Sun, S., & Park, U. (2021). Keypoint-based disentangled pose network for category-level 6-D object pose tracking. *IEEE Computer Graphics and Applications*, 42(5), 28–36.
- Tao, R., Gavves, E., & Smeulders, A. W. M. (2016). Siamese instance search for tracking. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 1420–1429). IEEE.
- Thangavel, J., Kokul, T., Ramanan, A., & Fernando, S. (2023). Transformers in single object tracking: An experimental survey. *arXiv preprint arXiv:2302.11867*.
- Van De Weijer, J., Schmid, C., Verbeek, J., & Larlus, D. (2009). Learning color names for real-world applications. *IEEE Transactions on Image Processing*, 18(7), 1512–1523.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., et al. (2017). Attention is all you need. In *Advances in neural information processing systems*, Vol. 30.
- Voigtlaender, P., Luiten, J., Torr, P. H., & Leibe, B. (2020). Siam r-cnn: Visual tracking by re-detection. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 6578–6588). IEEE.
- Wang, G., Luo, C., Xiong, Z., & Zeng, W. (2019). Spm-tracker: Series-parallel matching for real-time visual object tracking. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 3643–3652). IEEE.
- Wang, Q., Teng, Z., Xing, J., Gao, J., Hu, W., & Maybank, S. (2018). Learning attentions: Residual attentional siamese network for high performance online visual tracking. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 4854–4863).
- Wang, Q., Zhang, L., Bertinetto, L., Hu, W., & Torr, P. H. (2019). Fast online object tracking and segmentation: A unifying approach. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 1328–1338). IEEE.
- Wu, Y., Lim, J., & Yang, M.-H. (2015). Object tracking benchmark. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 37(9), 1834–1848.
- Wu, Q., Yang, T., Liu, Z., Wu, B., Shan, Y., & Chan, A. B. (2023). Dropmae: Masked autoencoders with spatial-attention dropout for tracking tasks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 14561–14571).
- Xing, D., Evangelidou, N., Tsoukalas, A., & Tzes, A. (2022). Siamese transformer pyramid networks for real-time UAV tracking. In *IEEE/CVF winter conference on applications of computer vision* (pp. 2139–2148).
- Xu, Y., Wang, Z., Li, Z., Yuan, Y., & Yu, G. (2020). Siamfc++: Towards robust and accurate visual tracking with target estimation guidelines. In *AAAI conference on artificial intelligence*, Vol. 34 (pp. 12549–12556).
- Xu, N., Yang, L., Fan, Y., Yang, J., Yue, D., Liang, Y., et al. (2018). Youtube-vos: Sequence-to-sequence video object segmentation. In *European conference on computer vision (ECCV)* (pp. 585–601).
- Yan, B., Peng, H., Fu, J., Wang, D., & Lu, H. (2021). Learning spatio-temporal transformer for visual tracking. In *International conference on computer vision (ICCV)* (pp. 10448–10457). IEEE.
- Yan, B., Wang, D., Lu, H., & Yang, X. (2020). Cooling-shrinking attack: Blinding the tracker with imperceptible noises. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 990–999). IEEE.
- Yan, B., Zhao, H., Wang, D., Lu, H., & Yang, X. (2019). ‘Skimming-perusal’ tracking: A framework for real-time and robust long-term tracking. In *International conference on computer vision (ICCV)* (pp. 2385–2393). IEEE.
- Yang, K., He, Z., Pei, W., Zhou, Z., Li, X., Yuan, D., et al. (2021). SiamCorners: Siamese corner networks for visual tracking. *IEEE Transactions on Multimedia*, 24, 1956–1967.
- Yang, H., Shao, L., Zheng, F., Wang, L., & Song, Z. (2011). Recent advances and trends in visual tracking: A review. *Neurocomputing*, 74(18), 3823–3831.
- Yao, Y., Wu, X., Zhang, L., Shan, S., & Zuo, W. (2018). Joint representation and truncated inference learning for correlation filter based tracking. In *European conference on computer vision (ECCV)* (pp. 552–567). Springer.
- Ye, B., Chang, H., Ma, B., Shan, S., & Chen, X. (2022). Joint feature learning and relation modeling for tracking: A one-stream framework. In *European conference on computer vision* (pp. 341–357). Springer.
- Yu, Y., Xiong, Y., Huang, W., & Scott, M. R. (2020). Deformable siamese attention networks for visual object tracking. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 6728–6737). IEEE.
- Zhang, Z., & Peng, H. (2019). Deeper and wider siamese networks for real-time visual tracking. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 4591–4600).
- Zhang, Z., Peng, H., Fu, J., Li, B., & Hu, W. (2020). Ocean: Object-aware anchor-free tracking. In *European conference on computer vision (ECCV)* (pp. 771–787). Springer.
- Zhang, Y., Wang, L., Qi, J., Wang, D., Feng, M., & Lu, H. (2018). Structured siamese network for real-time visual tracking. In *European conference on computer vision (ECCV)* (pp. 351–366). Springer-Verlag.
- Zhou, W., Wen, L., Zhang, L., Du, D., Luo, T., & Wu, Y. (2021). Siamcan: Real-time visual tracking based on siamese center-aware network. *IEEE Transactions on Image Processing*, 30, 3597–3609.
- Zhou, X., Zhuo, J., & Krahenbuhl, P. (2019). Bottom-up object detection by grouping extreme and center points. In *IEEE conference on computer vision and pattern recognition (CVPR)*. IEEE.
- Zhu, C., He, Y., & Savvides, M. (2019). Feature selective anchor-free module for single-shot object detection. In *IEEE conference on computer vision and pattern recognition (CVPR)* (pp. 840–849). IEEE.
- Zhu, Z., Wang, Q., Li, B., Wu, W., Yan, J., & Hu, W. (2018). Distractor-aware siamese networks for visual object tracking. In *European conference on computer vision (ECCV)* (pp. 103–119). Springer-Verlag.
- Zhu, Z., Wu, W., Zou, W., & Yan, J. (2018). End-to-end flow correlation tracking with spatial-temporal attention. In *IEEE conference on computer vision and pattern recognition (CVPR)*.